TEXTBOOK · FALL2 0 2 6

# Linear Algebra I

## Weeks 5–7

All questions and all formulas from each course file and assigned textbook sections — geometry and elimination, matrix operations and decompositions, transposes, permutations and vector spaces — each problem with $a$ complete step-by-step solution. Compiled for independent study and Midterm preparation

Tashkent Technological University · Fall 2026

Sources: course notes, lectures, $ISA$, practical sessions; $G$. Strang, Introduction to Linear Algebra $(4e)$ and Linear Algebra and its Applications $(4e)$

QUESTIONBANK · FORMULACATALOG · FULLSOLUTIONSOCTOBER2 0 2 6

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# 1 Course Map and How to Use This Document

| Week | Linear Algebra $I$ |
|---|---|
| 5 | Geometry and elimination: $row$ picture and column picture, Gaussian elimination, back substitution, elimination matrices $E$, permutation matrices $P$, connection with $rank, A = LU$(Strang §§2.1–2.3) |
| 6 | Matrix operations: multiplication rules and four interpretations of $AB$, special matrices, inverses and their derivation, elimination matrices, $LU/LDU$ decompositions, transpositions, permutations (Strang §§2.4–2.7) |
| 7 | Transpositions and permutations: transposition identities, symmetric matrices, six $3 \times 3$ permutations; vector spaces and subspaces, column space $C(A), null$ space $N(A)$(Strang §§2.7, 3.1) |

This guide continues the official collection "Linear Algebra $I$ — Weeks 1–4" and covers weeks 5–7 of the fall semester 2026. $It$ collects every question and every formula from each course file and the assigned textbook sections — each problem with $a$ complete step-by-step solution, as in the main collection.

Course files covered (weeks 5–7): lectures; practical sessions and in-class problems with solutions; Independent Study Assignments $(ISA)$; the expanded week 5 Beamer theory notes; EXTRA material (week 6 review with examples). Textbooks: $G$. Strang, Introduction to Linear Algebra $(4thed.)$, with the parallel companion Linear Algebra and Its Applications $(4thed.)$.

How each week is organised. First comes the formula catalog: all definitions, theorems and rules of the week with explanations. Then, for every source file, $a$ "questions and complete solutions" section: in-class problems, the practical session, Independent Study Assignments $(ISA)$ and extra material. Problems are reproduced verbatim; solutions are complete, step by step, with checks.

Notation. Matrices are written $row$ by $row$: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — first $rowab$, second $rowcd$; $a$ column vector is written $\begin{bmatrix} x \\ y \end{bmatrix}$, $arow$ vector $\begin{bmatrix} x & y \end{bmatrix}$. Powers and indices are written with ^{} and _: $A^{-1}, A^T, a_{11}$. Weeks 5–7 close the first half of the course: elimination and matrix operations, the decompositions $A = LU = LDU = PA$, transposes, permutations and vector spaces. Together with the weeks 1–4 material they form the base assumed known for the week 8 Midterm.

# 2 Week 5 — Geometry and Elimination (Linear AlgebraI)

Week 5 of Linear Algebra $I(TT2026)$ connects the geometry of the equation $Ax = b$ with the algorithm for solving it—Gaussian elimination. Sources: Strang, Introduction to Linear Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, 4th ed., §§1.2–1.4; Beamer lecture notes "Geometry and Elimination" (80 minutes lecture, 90 minutes problems). Weekly plan: review of $rank$ and inverses $(10min), row$ and column

pictures $(20min)$, elimination and back substitution $(25min)$, elimination matrices $E$ and permutation matrices $P(20min)$, summary $(5min)$. Objectives: to see $Ax = b$ as the intersection of lines and planes $(row$ picture) and as $a$ linear combination of columns (column picture); to perform elimination with multipliers and back substitution; to represent $row$ operations with matrices $E$, cancel them with matrices $E^{-1}$, and swap equations with matrices $P$. Notation: matrices are written $row$-wise — $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (first $rowab$, second $cd$); column vector $\begin{bmatrix} x \\ y \end{bmatrix}$; augmented matrix $[A|b]$.

## 2.1 Linear Algebra I, week 5: formula catalog

[Source: Beamer lecture notes week 5 "Geometry and Elimination" (Linear Algebra $I, TT2026$): Strang, Introduction to Linear

Algebra, 4th ed., §§2.1–2.3; Linear Algebra and Its Applications, §§1.2–1.4]

### 2.6.1 Connection with $rank$: $n$ pivots mean invertibility

Review: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $rank 1$ and no inverse; $Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions, while $Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ has none. Key fact: for an $n \times n$ square matrix, having $n$ pivots is equivalent to invertibility—then $Ax = b$ has $a$ unique solution for every $b$; the number of pivots found by elimination is the $rank$. Both algorithms act on $[A|b]$: Gauss–Jordan reduces the matrix to reduced echelon form; Gaussian elimination stops at upper triangular $U$ and completes with back substitution.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} : rankA = 1, A^{-1} doesnotexist;$$

$$Ax = \begin{bmatrix} 3 \\ 6 \end{bmatrix} — infinitelymanysolutions, Ax = \begin{bmatrix} 3 \\ 7 \end{bmatrix} — nosolution$$

### 2.6.2 Row picture: rows as lines and planes

$$2x - y = 1, x + y = 5;$$
$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Each $row$ is one geometric constraint: in 2D lines, in 3D planes; the solution is their common point. The example lines intersect at $(x, y) = (2, 3)$; three planes in 3D:

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} — planesintersectat(u, v, w) = (1, 1, 2)$$

### 2.6.3 Column picture: $Ax$ as $a$ linear combination of columns

$$x \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix};$$
$$check: 2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

**Strang emphasizes: $Ax$ is $a$ linear combination of the columns of $A$—we** choose coefficients so that $x_1 \cdot a_1 + \ldots + x_n \cdot a_n = b$. Three-dimensional version (weights (1, 1, 2)):

$$u \cdot \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \cdot \begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + w \cdot \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Both pictures are the same equation $Ax = b$: rows are intersections (equations), columns are linear combinations (vectors).

### 2.6.4 Three possible outcomes

- • One solution—independent constraints. • Infinitely many solutions: $x + y = 2$ and 2x + 2y $= 4$—the second equation is

twice the first, the lines coincide.

- • $No$ solutions: $x + y = 2$ and 2x + 2y $= 5$—the left sides are proportional, the

right sides are not, the lines are parallel.

$$eliminationrevealsoutcome: [00\ldots0|0] \to 0 = 0—infinitelymanysolutions;$$
$$[00\ldots0|c], c \neq 0,$$

$$\to 0 = c—nosolutions$$

### 2.6.5 Pivot, multiplier, and forward elimination

$$\lambda_{ij} = \frac{a_{ij}}{a_{jj}}, operationR_i \leftarrow R_i - \lambda_{ij} \cdot R_j$$

$To$ eliminate element $a_{ij}$ below $a$ nonzero pivot $a_{jj}(the$ multiplier is denoted $\lambda_{ij}$ here), subtract $rowj$ multiplied by $\lambda_{ij}$ from $rowi$. Running example: $\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]$; pivot 2, multipliers $\lambda_{21} = \frac{4}{2} = 2$ and $\lambda_{31} = -\frac{2}{2} = -1$:

$$R_2 \leftarrow R_2 - 2R_1,$$
$$R_3 \leftarrow R_3 + R_1 : \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right]$$

Second pivot $-8$, multiplier $\lambda_{32} = 8/(-8) = -1$, operation $R_3 \leftarrow R_3 + R_2$. Triangular system and result of forward elimination:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right],$$
$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}, pivotelements 2, -8, 1$$

### 2.6.6 Back substitution

$$w = 2; -8v - 2w = -12 \rightarrow v = 1;$$
$$2u + v + w = 5 \rightarrow u = 1; x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

The triangular system is solved from bottom up: the last equation contains only $w$, the second to last contains $v$ and $w$. The answer is verified by substitution into the original equation: $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$. $Arow$ operation is reversible ($add 2R_1$ back), so elimination preserves solutions; but the operation is applied to both left and right sides, otherwise the equations are not equivalent.

### 2.6.7 Elimination matrices $E$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$
$$E_{21} \cdot A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix}, E_{21} \cdot b = \begin{bmatrix} 5 \\ -12 \\ 9 \end{bmatrix}$$

Left multiplication performs $arow$ operation, and the right side is transformed by the same matrix: $Ax = b$ becomes $(E_{21}A)x = E_{21}b$. Full pass to $U(the$ rightmost multiplier acts first—order of multiplication matters):

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix},$$

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$$

Undo step: subtraction is replaced by addition—for $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ the inverse is $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ and $E^{-1} \cdot E = I$.

### 2.6.8 Zero pivot: permutation matrices $P$

$$\begin{bmatrix} 0 & 2 & | & 4 \\ 3 & 1 & | & 7 \end{bmatrix} \to \begin{bmatrix} 3 & 1 & | & 7 \\ 0 & 2 & | & 4 \end{bmatrix} (rowexchange),$$

$$theny = 2, x = \frac{5}{3}$$

$A$ zero pivot does not necessarily mean singularity: if there is $a$ nonzero element below, rows are swapped. The swap is performed by $a$ permutation matrix, and it also transforms the right side:

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$P \cdot \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, P \cdot \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}, equivalentsystemPAx = Pb$$

Properties of $P$: one 1 in each $row$ and column, rest zeros; $P^{-1} = P^T$; for $a$ single swap $P^2 = I$; swapping rows 1 and 3 — $P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; left multiplication changes equations, not unknowns. Elimination failures: temporary — zero pivot with nonzero element below (swap saves); permanent — no available pivot in column (singularity: no solution or infinitely many).

### 2.6.9 Looking ahead: $A = LU$ and $PA = LU$

$$withoutrowexchangesA = LU(U—uppertriangular,$$
$$Lstoreseliminationfactors); withexchanges$$

$$PA = LU$$

Week summary: rows — intersections, columns — combinations; elimination — zeros below pivots plus back substitution; $E$ — $row$ replacement, $P$ — $row$ swap. $LU$ decomposition ($A = LU, PA = LU$) is built in week 6.

## 2.2 Linear Algebra I, week 5: in-class problems — questions andcomplete solutions

[Source: Week 5 Class Problems (Projection), problems 1–15($law5$classg$01$–$g11$ and text layer pp. 23–31)]

**Exercise 1. Row picture and column picture. Solve the system** $\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ **and describe both geometric pictures.**

Solution. Add equations: 3x $= 6, x = 2$, then $y = 5 - 2 = 3$. Lines 2x $- y = 1$ and $x + y = 5$ intersect at (2, 3); by columns: $2 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 & & 3 \\ 2 & + & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. Answer: $x = 2, y = 3$.

**Exercise 2. Three outcomes. Classify the system** $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ c \end{bmatrix}$ **for** $c = 4$ **and** $c = 5$**.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $row$ [$00|c - 4$]. When $c = 4$ this is $0 = 0$: $x + y = 2$ remains, infinitely many solutions — $\begin{bmatrix} 2 & t \\ t & \end{bmatrix}$. When $c = 5$ this is $0 = 1$ — contradiction, no solution.

**Exercise 3. Find the multiplier. Eliminate below the first pivot in the augmented matrix** $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 8 & 13 & 29 \end{array}\right]$ **and find both pivots.**

Solution. Multiplier $\lambda_{21} = \frac{8}{2} = 4$. $R_2 \leftarrow R_2 - 4R_1$: [8, 13 | 29] $-4\cdot$[2, 3 | 7] = [0, 1 | 1]. Result: $\left[\begin{array}{cc|c} 2 & 3 & 7 \\ 0 & 1 & 1 \end{array}\right]$; pivots are 2 and 1.

**Exercise 4. Back substitution. Solve the system** $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 8 \end{bmatrix}$**.**

Solution. Bottom up: 4z $= 8, z = 2$; 3y $+2 \cdot 2 = 7, y = 1$; 2x $+1 - 2 = 3, x = 2$. Answer: $x = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$. Check: $2 \cdot 2 + 1 - 2 = 3, 3 + 4 = 7, 8 = 8$ — correct.

**Exercise 5. Full elimination. Solve the system**

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 10 \\ 1 & 2 & 3 & 14 \end{array}\right]$$

**by elimination.**

Solution. $R_2 \leftarrow R_2 - 2R_1 \rightarrow$ [0 1 −1 | −2]; $R_3 \leftarrow R_3 - R_1 \rightarrow$ [0 1 2 | 8]; $R_3 \leftarrow R_3 - R_2 \rightarrow$ [0 0 3 | 10], so 3z $= 10$. Back substitution: $z = \frac{10}{3}$; $y - z = -2 \rightarrow y = \frac{4}{3}$; $x + y + z = 6 \rightarrow x = \frac{4}{3}$. Answer: [4/3; 4/3; 10/3]. Check: rows give 6, 10, 14 — correct.

**Exercise 6. Identify elementary matrix. Find $E$ that performs $R_3 \leftarrow R_3 - 4R_1$ on $a 3 \times 3$ matrix.**

Solution. $E$ is identity matrix with −4 in position (3, 1): $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; $row3$ of $EA$ equals $(row3) - 4 \cdot (row1)$, rows 1–2 unchanged. Inverse operation is addition: $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$ with $+4$ in position (3, 1).

**Exercise 7. $EA$ multiplication. Let $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$. Compute $EA$ and explain the operation performed.**

Solution. Row 1 of product: $\begin{bmatrix} 3 & 1 \end{bmatrix}$; $row2$: $-2 \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 3 \end{bmatrix}$. $SoEA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$: $E$ subtracts twice $row1$ from $row2$ — elimination with multiplier $\frac{6}{3} = 2$.

**Exercise 8. Transformed right side. For $A = \begin{bmatrix} 3 & 1 \\ 6 & 5 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 17 \end{bmatrix}$ use $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ to solve the system.**

Solution. $Eb = \begin{bmatrix} 4 \\ -8 + 17 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; together with $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ (exercise 7) we get 3y $= 9, y = 3$ and 3x $+3 = 4, x = \frac{1}{3}$. Check in original system: $3 \cdot (\frac{1}{3}) + 3 = 4, 6 \cdot (\frac{1}{3}) + 5 \cdot 3 = 17$ — correct.

**Exercise 9. Zero pivot. Solve the system $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ by $row$ swap; find $P$.**

Solution. Zero in position (1, 1) is temporary failure: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$. Then 2y $= 4, y = 2$; 3x $+y = 7, x = \frac{5}{3}$. Check: $2 \cdot 2 = 4, 3 \cdot (\frac{5}{3}) + 2 = 7$ — correct; zero pivot didn'$t$ mean singularity.

**Exercise 10. Three-$row$ permutation. Find $P$ that swaps rows 1 and 3 of matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$; compute $PA$.**

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — rows of identity matrix in order 3, 2, 1; $row1$ of $PA$ equals

$row3$ of matrix $A$ and so on: $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. Swap $1 \leftrightarrow 3$ is involution: $P^{-1} = P$ and $P^2 = I$(consistent with $P^{-1} = P^T$).

**Exercise 11. Singular but consistent. Solve the system** $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$**.**

Solution. Row 2 with the right-hand side $(6 = 2 \cdot 3)$ is twice $row1$, so $R_2 \leftarrow R_2 - 2R_1$ gives $0 = 0$. $We$ are left with $x+$ 2y $+z = 3$ and $y + z = 2$; free variable $z = t$: $y = 2 - t, x + 2(2 - t) + t = 3 \rightarrow x = t - 1$. Answer: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$ — infinitely many solutions.

Verification for $t = 1$: $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ gives 3, 6, 2 — correct.

**Exercise 12. Degenerate and inconsistent. Replace the right-hand side in the previous system with** $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$**.** $Is$ **the system consistent?**

Solution. Now $7 \neq 2 \cdot 3$. $R_2 \leftarrow R_2 - 2R_1$ gives the $row$ [0 0 0 | 7 − 6] = [0 0 0 | 1], that is, $0 = 1$ — $a$ contradiction. The system is inconsistent: the right-hand side violates the necessary condition $b_2 = 2b_1$.

**Exercise 13. Column interpretation of consistency. For** $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$**, which vectors** $b =$ **[**$b_1$**;** $b_2$**] admit** $a$ **solution** $Ax = b$**?**

Solution. The second $row$ is twice the first: after $R_2 \leftarrow R_2 - 2R_1$ we have $[00|b_2 - 2b_1]$, so consistency requires $b_2 = 2b_1$. The columns $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are multiples of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so $Ax$ always lies on the line $b_2 = 2b_1$. Answer: $b = [b_1;\ 2b_1]$ — $b$ must lie on the line through the columns of $A$.

**Exercise 14. Order of matrix operations. Let** $E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ **and** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$**. Compare** $EP$ **and** $PE$**.**

Solution. $EP$: $row1 = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $row2 = -2 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} =$ [1, −2], $i.e.$ $EP = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$. $PE$: $row1 = 1 \cdot \begin{bmatrix} -2 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \end{bmatrix}$, $row2 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $i.e.$ $PE = \begin{bmatrix} -2 & 1 \\ 1 & 0 \end{bmatrix}$. The products are different: $row$ operations, in general, do not commute.

**Exercise 15. Challenge: construct** $a$ **system. Build** $a 3 \times 3$ **system requiring** $row$

**exchange, with three pivots and solution** $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$**.**

Solution. $A$ zero in position (1, 1) with $a$ nonzero element below: $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; set the right-hand side by the column picture: $b = A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$. Swapping rows 1 and $2 (P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix})$ turns $A$ into the identity matrix: pivots 1, 1, 1, solution $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ — correct.

## 2.3 Linear Algebra I, week 5: practical session — questions andcomplete solutions

[Source: Week 5 Practical Session problems with solutions (Week$5TA$ Problems Projection), problems 1–15; matrices restored
from the text layer]

**Exercise 1. System of two lines. Solve the system** $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$**.**

Solution. $R_2 \leftarrow R_2 - 3R_1$: $(1-6)y = 8-21$, that is, $-5y = -13, y = \frac{13}{5}$; then $x = 7-2\cdot(\frac{13}{5}) = \frac{9}{5}$. Answer: (9/5, 13/5). Verification: $\frac{9}{5} + \frac{26}{5} = 7$, $\frac{27}{5} + \frac{13}{5} = 8$ — correct.

**Exercise 2. Parallel or coincident? Classify the system** $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ c \end{bmatrix}$ **for** $c = 3$ **and** $c = 4$**.**

Solution. Row 1 is twice $row2$ on the left; consistency requires the same on the right: $6 = 2c$, that is, $c = 3$. For $c = 3$, we are left with one equation $x + 2y = 3$ — infinitely many solutions $\begin{bmatrix} 3 & 2t \\ t \end{bmatrix}$ ($the$ lines coincide). For $c = 4$, doubling the second equation gives $2x + 4y = 8 \neq 6$ — no solutions ($the$ lines are parallel).

**Exercise 3. One elimination step. Eliminate below the first pivot in the matrix** $\begin{bmatrix} 3 & 2 \\ 9 & 8 \end{bmatrix}$**.**

Solution. The multiplier $\lambda_{21} = \frac{9}{3} = 3$; $R_2 \leftarrow R_2 - 3R_1$: [9 − 9, 8 − 6] = [0, 2]. Result: $\begin{bmatrix} 3 & 2 \\ 0 & 2 \end{bmatrix}$; pivots 3 and 2.

**Exercise 4. Triangular system. Solve** $\begin{bmatrix}1&2&1\\0&2&-2\\0&0&3\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}8\\0\\6\end{bmatrix}$**.**

Solution. From bottom up: 3z $=6, z=2$; 2y $-2\cdot 2=0, y=2$; $x+4+2=8, x=2$. Answer: $\begin{bmatrix}2\\2\\2\end{bmatrix}$. Verification: $2+4+2=8, 4-4=0, 3\cdot 2=6$ — correct.

**Exercise 5. Complete system with three unknowns. Solve the system**

$$\left[\begin{array}{ccc|c}1&1&0&3\\2&3&1&9\\1&0&1&4\end{array}\right]$$

**.**

Solution. $R_2\leftarrow R_2-2R_1\rightarrow$ [0 1 1 | 3]; $R_3\leftarrow R_3-R_1\rightarrow$ [0 −1 1 | 1]; $R_3\leftarrow R_3+R_2\rightarrow$ [0 0 2 | 4], that is, 2z $=4, z=2$. Backwards: $y+z=3\rightarrow y=1$; $x+y=3\rightarrow x=2$. Answer: $\begin{bmatrix}2\\1\\2\end{bmatrix}$. Verification: $2+1=3, 4+3+2=9, 2+2=4$ — correct.

**Exercise 6. Construct** $E_{32}$**. Write the** $3\times 3$ **matrix for the operation** $R_3\leftarrow R_3+5R_2$**.**

Solution. $E_{32}$ is the identity matrix with 5 in position (3, 2): $E_{32}=\begin{bmatrix}1&0&0\\0&1&0\\0&5&1\end{bmatrix}$; rows 1–2 are unchanged, and $row3$ of the product equals $(row3)+5\cdot(row2)$.

**Exercise 7. Reverse the operation. Find the inverse matrix for** $E=\begin{bmatrix}1&0&0\\0&1&0\\0&-3&1\end{bmatrix}$**.**

Solution. $E$ implements $R_3\leftarrow R_3-3R_2$(subtracting three times $row2$); the inverse operation is addition: $E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\0&3&1\end{bmatrix}$ ($the$ sign changed). Verification: $E^{-1}\cdot E=I$.

**Exercise 8. Transform the right-hand side. Apply** $E=\begin{bmatrix}1&0&0\\-2&1&0\\0&0&1\end{bmatrix}$ **to** $b=\begin{bmatrix}3\\5\\7\end{bmatrix}$**.**

Solution. $Eb=$ [3; $5-2\cdot 3$; 7] $=\begin{bmatrix}3\\-1\\7\end{bmatrix}$: the second component becomes $5-2\cdot 3=-1$. $E$ eliminates $x$ from the second equation with multiplier 2 — the right-hand side is transformed by the same operation.

**Exercise 9. Row exchange matrix. Find** $P$ **that swaps rows 2 and 3 of** $a3\times 3$ **matrix.**

Solution. $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is the identity matrix with rows 2 and 3 swapped. $A$ double swap returns the original order: $P^2 = I$, so $P^{-1} = P$(consistent with $P^{-1} = P^T$); left multiplication swaps equations, not unknowns.

**Exercise 10. System requiring exchange. Solve the system** $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \end{bmatrix}$**.**

Solution. The pivot in position (1, 1) is zero — swap rows: 2x + 3y $= 14$ and $y = 4$. Then 2x $+12 = 14, x = 1$. Answer: (1, 4). Check: $0 \cdot 1 + 4 = 4, 2 + 12 = 14$ — correct.

**Exercise 11. Free variable. Solve the system** $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix}$**.**

Solution. The third $row$ is $0 = 0$, giving no constraints; there are two pivots and three unknowns. Let $z = t$: $y + t = 3 \to y = 3 - t$; $x + (3 - t) + t = 4 \to x = 1$. Answer: $\begin{bmatrix} 1 & \\ 3 & t \\ t & \end{bmatrix}$.

Check at $t = 0$: $\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ gives 4, 3, 0 — correct.

**Exercise 12. Inconsistent** $row$**. Explain why the system**

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

**has no solutions.**

Solution. The last $row$ reads as $0 \cdot x + 0 \cdot y + 0 \cdot z = 1, i.e., 0 = 1$ — an impossible equality: $a$ zero left-hand side cannot give $a$ nonzero right-hand side for any $x, y, z$. The system is inconsistent.

**Exercise 13. Compare** $EP$ **and** $PE$**. For** $E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ **and** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **compute both products.**

Solution. $EP$: $row1 = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $row2 = 4 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \end{bmatrix} =$ [1, 4], $i.e.$, $EP = \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}$. $PE$: $row1 = 1 \cdot \begin{bmatrix} 4 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \end{bmatrix}$, $row2 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $i.e.$, $PE = \begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$. The products are different: ”first elimination, then exchange” and ”first exchange, then elimination” give different matrices.

**Exercise 14. Column space test. Can the system** $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ **be solved?**

Solution. Both columns have the second component twice the first: $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}3\\6\end{bmatrix} = 3\cdot\begin{bmatrix}1\\2\end{bmatrix}$; any combination $x\cdot\begin{bmatrix}1\\2\end{bmatrix}+y\cdot\begin{bmatrix}3\\6\end{bmatrix}$ has the form $\begin{bmatrix}s\\2s\end{bmatrix}$. For $b=\begin{bmatrix}2\\5\end{bmatrix}$ we have $5\neq 2\cdot 2=4$ — $b$ is not in the column space. Answer: no (doubling the first equation gives 2x + 6y $= 4 \neq 5$ — the lines are parallel).

**Exercise 15. Challenge: construct $a$ permutation. Construct $P$ that maps rows $(R_1, R_2, R_3)$ to $(R_3, R_1, R_2)$.**

Solution. New $row1$ is old $R_3$, new 2 is old $R_1$, new 3 is old $R_2$, so the rows of $P$ are the rows of the identity matrix in order 3, 1, 2: $P=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$. Check on rows $\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}$: $P\cdot A=\begin{bmatrix}7&8&9\\1&2&3\\4&5&6\end{bmatrix}$. Left multiplication permutes rows; $P^3=I$.

## 2.4 Linear Algebra I, week 5: ISA exercises — questions andcomplete solutions

[Source: Week 5 Independent Study Assignments, practices 1–15; no solutions file provided — complete solutions with verification

are given below]

**Practice 1. Row picture and column picture. Solve the system $\begin{bmatrix}2&-1\\1&1\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}1\\5\end{bmatrix}$. Describe both geometric pictures. Show each elimination step and verify the answer.**

Solution. Add the equations: 3x $= 6, x = 2, y = 3$. Check: $2\cdot 2-3=1, 2+3=5$. The lines 2x $-y=1$ and $x+y=5$ intersect at (2, 3); in columns: $2\cdot\begin{bmatrix}2\\1\end{bmatrix}+3\cdot\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}1\\5\end{bmatrix}$. Answer: $x=2, y=3$.

**Practice 2. Three outcomes. Classify the system $\begin{bmatrix}1&1\\2&2\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}2\\c\end{bmatrix}$ for $c=4$ and $c=5$. Show each elimination step and verify the answer.**

Solution. $R_2\leftarrow R_2-2R_1$ gives $[00|c-4]$. For $c=4$: $0=0$, solutions $\begin{bmatrix}2&t\\t&\end{bmatrix}$ (check: $(2-t)+t=2, 2(2-t)+$ 2t $=4$). For $c=5$: $0=1$ — no solutions.

**Practice 3. Find the multiplier. Eliminate below the first pivot in $\left[\begin{array}{cc|c}2&3&7\\8&13&29\end{array}\right]$; find both pivots. Show each step and verify the answer.**

Solution. $\lambda_{21}=\frac{8}{2}=4$; $R_2\leftarrow R_2-4R_1$: [8, 13 | 29] $-4\cdot$[2, 3 | 7] = [0, 1 | 1]. Result: $\left[\begin{array}{cc|c}2&3&7\\0&1&1\end{array}\right]$, pivots are 2 and 1. Check by invertibility: [0, 1 | 1] $+4\cdot$[2, 3 | 7] = [8, 13

| 29].

**Practice 4. Back substitution. Solve** $\begin{bmatrix}2&1&-1\\0&3&2\\0&0&4\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$**. Show each step and verify the answer.**

Solution. From bottom up: $z=\frac{8}{4}=2$; 3y $+2\cdot 2=7, y=1$; 2x $+1-2=3, x=2$. Verification: $\begin{bmatrix}2&1&-1\\0&3&2\\0&0&4\end{bmatrix}\cdot\begin{bmatrix}2\\1\\2\end{bmatrix}=\begin{bmatrix}3\\7\\8\end{bmatrix}$. Answer: $x=\begin{bmatrix}2\\1\\2\end{bmatrix}$.

**Practice 5. Complete elimination. Solve the system**

$$\left[\begin{array}{ccc|c}1&1&1&6\\2&3&1&10\\1&2&3&14\end{array}\right]$$

**by elimination. Show each step and verify the answer.**

Solution. $R_2\leftarrow R_2-2R_1$ and $R_3\leftarrow R_3-R_1$ give $\left[\begin{array}{ccc|c}1&1&1&6\\0&1&-1&-2\\0&1&2&8\end{array}\right]$; then $R_3\leftarrow R_3-R_2$ gives [0 0 3 | 10]. Backwards: $z=\frac{10}{3}, y=z-2=\frac{4}{3}, x=6-y-z=\frac{4}{3}$. Verification: the rows give 6, 10, 14 — correct. Answer: [4/3; 4/3; 10/3].

**Practice 6. Identify the elementary matrix. Find** $E$ **that implements** $R_3\leftarrow R_3-4R_1$ **on** $a3\times 3$ **matrix. Show each step and verify the answer.**

Solution. $E=\begin{bmatrix}1&0&0\\0&1&0\\-4&0&1\end{bmatrix}$ — the identity with −4 in position (3, 1); the third $row$ of $EA$ is $(row3)-4\cdot(row1)$. Verification on $\begin{bmatrix}1&1&1\\2&3&1\\1&2&3\end{bmatrix}$: $row3$ is $\begin{bmatrix}1&2&3\end{bmatrix}-4\cdot\begin{bmatrix}1&1&1\end{bmatrix}=\begin{bmatrix}-3&-2&-1\end{bmatrix}$. Inverse: $E^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\4&0&1\end{bmatrix}$.

**Practice 7. Multiplication** $EA$**. Let** $E=\begin{bmatrix}1&0\\-2&1\end{bmatrix}, A=\begin{bmatrix}3&1\\6&5\end{bmatrix}$**. Compute** $EA$ **and explain the operation. Show each step and verify the answer.**

Solution. Row 1: $\begin{bmatrix}3&1\end{bmatrix}$; $row2$: $\begin{bmatrix}6&5\end{bmatrix}-2\cdot\begin{bmatrix}3&1\end{bmatrix}=\begin{bmatrix}0&3\end{bmatrix}$; $EA=\begin{bmatrix}3&1\\0&3\end{bmatrix}$ — subtracting twice $row1$ from $row2$(multiplier 2). Verification: $E^{-1}=\begin{bmatrix}1&0\\2&1\end{bmatrix}$ returns $A$.

**Practice 8. Transformed right-hand side. For** $A=\begin{bmatrix}3&1\\6&5\end{bmatrix}$ **and** $b=\begin{bmatrix}4\\17\end{bmatrix}$ **use**

$E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ **to solve the system. Show each step and verify the answer.**

Solution. $Eb = [4;\ -2 \cdot 4 + 17] = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$; $EA = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$; then 3y $= 9 \to y = 3$ and 3x $+3 = 4 \to x = \frac{1}{3}$. Verification in the original system: $3 \cdot (\frac{1}{3}) + 3 = 4, 6 \cdot (\frac{1}{3}) + 5 \cdot 3 = 17$ — correct.

**Practice 9. Zero pivot element. Solve** $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$ **by** $row$ **exchange; find** $P$**. Show each step and verify the answer.**

Solution. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; $PA = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}, Pb = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$; 2y $= 4 \to y = 2$, 3x $+2 = 7 \to x = \frac{5}{3}$. Verification: $0 \cdot (\frac{5}{3}) + 2 \cdot 2 = 4, 3 \cdot (\frac{5}{3}) + 2 = 7$ — correct.

**Practice 10. Permutation of three rows. Find** $P$ **that swaps rows 1 and 3 of matrix** $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 4 & 0 & 2 \end{bmatrix}$**; compute** $PA$**. Show each step and verify the answer.**

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; $row1$ of $PA = row3$ of $A$ and $row3$ of $PA = row1$ of $A$, so $PA = \begin{bmatrix} 4 & 0 & 2 \\ 0 & 1 & 3 \\ 1 & 2 & 0 \end{bmatrix}$. Verification: $P^2 = I, i.e.,\ P \cdot PA = A$; $P^{-1} = P = P^T$.

**Practice 11. Degenerate but consistent. Solve** $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$**. Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $[000|6 - 2 \cdot 3] = [0\ 0\ 0 \mid 0]$ — $arow$ disappeared, the system is consistent but degenerate. Remaining: $x+$ 2y $+z = 3$ and $y + z = 2$; $z = t$: $y = 2 - t, x = t - 1$. Answer: $\begin{bmatrix} t & 1 \\ 2 & t \\ t & \end{bmatrix}$. Verification at $t = 2$: $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ gives 3, 6, 2 — correct.

**Practice 12. Degenerate and inconsistent.** $In$ **the previous system, replace the right-hand side with** $b = \begin{bmatrix} 3 \\ 7 \\ 2 \end{bmatrix}$**.** $Is$ **the system consistent? Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ gives $[000|7 - 2 \cdot 3] = [0\ 0\ 0 \mid 1]$, $i.e.,\ 0 = 1$ — contradiction. The system is inconsistent: the necessary condition $b_2 = 2b_1$ is violated $(7 \neq 6)$.

**Practice 13. Column interpretation of consistency. For** $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$**, which**

**vectors $b = [b_1; b_2]$ admit $a$ solution to $Ax = b$? Show each step and verify the answer.**

Solution. $R_2 \leftarrow R_2 - 2R_1$ leaves $[12|b_1;\ 00|b_2 - 2b_1]$: consistency requires $b_2 = 2b_1$. The columns $\begin{bmatrix}1\\2\end{bmatrix}$ and $\begin{bmatrix}2\\4\end{bmatrix} = 2\cdot\begin{bmatrix}1\\2\end{bmatrix}$ are multiples of $\begin{bmatrix}1\\2\end{bmatrix}$, so $Ax$ always lies on the line $b_2 = 2b_1$. Answer: $b = [b_1;\ 2b_1]$. Verification: $\begin{bmatrix}4\\8\end{bmatrix}$ is solved $(x = 4, y = 0)$, $\begin{bmatrix}4\\9\end{bmatrix}$ is not.

**Practice 14. Order of matrix operations. Let $E = \begin{bmatrix}1 & 0\\-2 & 1\end{bmatrix}$ and $P = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$. Compare $EP$ and $PE$. Show each step and verify the answer.**

Solution. $EP$: rows $\begin{bmatrix}0 & 1\end{bmatrix}$ and $[1,\ -2]$, *i.e.*, $EP = \begin{bmatrix}0 & 1\\1 & -2\end{bmatrix}$. $PE$: rows $\begin{bmatrix}-2 & 1\end{bmatrix}$ and $\begin{bmatrix}1 & 0\end{bmatrix}$, *i.e.*, $PE = \begin{bmatrix}-2 & 1\\1 & 0\end{bmatrix}$. The products are different — *row* operations do not commute: in $EP$ elimination acts on the permuted rows, in $PE$ on the original ones.

**Practice 15. Challenge: construct $a$ system. Build $a 3\times 3$ system that requires *row* exchange, with three pivots and solution $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Show each step and verify the answer.**

Solution. $A = \begin{bmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix}$ — zero in position (1, 1), with $a$ one below; $b = A\cdot\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$. Swapping rows 1 and $2 (P = \begin{bmatrix}0 & 1 & 0\\1 & 0 & 0\\0 & 0 & 1\end{bmatrix})$ gives the identity matrix: pivots 1, 1, 1, solution $x = \begin{bmatrix}1\\2\\3\end{bmatrix}$. Verification: $A\cdot\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix}$; without exchange, the first column is not eliminated.

# 3 Week 6 — Matrix Operations, Inverses andDecompositions (Linear Algebra I)

## 3.1 Linear Algebra I, week 6: formula catalog

[Source: Week 6 Lecture presentation (Strang, Introduction to Linear Algebra, 4th ed., §§2.4–2.7; companion: Lay, §§1.4–1.6)]

### 3.5.1 Review of week 5: two pictures of the system and elimination

Row picture: equations are intersecting lines or planes. Column picture: $Ax = x_1\cdot a_1 + ... + x_n\cdot a_n = b$. Example: $\begin{bmatrix}2 & 1\\1 & 1\end{bmatrix}\cdot\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}5\\3\end{bmatrix}$ gives $(x, y) = (2, 1)$; both pictures describe

the same system. Elimination: with pivot $a_{kk} \neq 0$, the multiplier $m_{ik} = a_{ik}/a_{kk}$ is taken and $R_i \leftarrow R_i - m_{ik} \cdot R_k$ is performed; thus $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, then back substitution solves the triangular system. Degeneracy: $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$ has infinitely many solutions, while changing 6 to 7 gives the contradiction $0 = 1$. For square matrices, $a$ full set of pivots means $a$ unique solution for any right-hand side.

### 3.5.2 Multiplication rules and four interpretations of $AB$

$$A(m \times n), B(n \times p) \rightarrow AB(m \times p), (AB)_{ij} = \sum_k a_{ik} \cdot b_{kj}$$

Example: $\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 10 & 4 \end{bmatrix}$. Four views of $AB$: (1) element-wise — $rowi$ of matrix $A$ is multiplied by column $j$ of matrix $B$; (2) column-wise — $AB = [Ab_1 ... Ab_p]$; $(3) row$-wise — each $row$ of $AB$ is $a$ combination of rows of $B$; (4) outer products — $AB = \sum_k a_k \cdot r_k$(column $k$ of matrix $A$ times $rowk$ of matrix $B$).

$$(AB)C = A(BC), A(B + C) = AB + AC, (A + B)C = AC + BC, AI = IA = A$$

$$generallyspeakingAB \neq BA; (A + B)^2 = A^2 + AB + BA + B^2 \neq A^2 + 2AB + B^2$$

$An$ explicit example of non-commutativity: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ give $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, but $BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ — horizontal and vertical shears depend on the order of application.

### 3.5.3 Special matrices and elimination matrices

The identity matrix $I$ leaves vectors unchanged; diagonal $D = diag(d_1, ..., d_n)$ scales coordinates. Upper triangular matrices have zeros below the diagonal, lower triangular matrices have zeros above it; triangular systems are solved by forward/backward substitution. The elimination operation is itself $a$ matrix: $R_2 \leftarrow R_2 - 3R_1$ corresponds to $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$, and $E \cdot \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$; the transformed system is EAx $= Eb$, and $E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ restores the original rows.

### 3.5.4 The inverse matrix: definition, criteria, rules

$$AA^{-1} = A^{-1}A = I \rightarrow Ax = bhassolutionx = A^{-1}bforanyb$$

The inverse reverses the linear transformation. $IfAx = 0$ for nonzero $x$, the inverse does not exist. For $a$ square $A$, the following are equivalent: the inverse exists $\Leftrightarrow$ all pivots are nonzero (after possible exchanges) $\Leftrightarrow$ columns are independent $\Leftrightarrow Ax = 0$ only when $x = 0 \Leftrightarrow Ax = b$ has $a$ unique solution for any $b$.

$$(AB)^{-1} = B^{-1}A^{-1},$$
$$(A^T)^{-1} = (A^{-1})^T, (A^{-1})^{-1} = A, (A^k)^{-1} = (A^{-1})^k$$

The order of factors in the inverse of $a$ product is reversed: $(AB)(B^{-1}A^{-1}) = I$.

### 3.5.5 The inverse of $2 \times 2$ and the Gauss-Jordan method

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} d & b \\ c & a \end{bmatrix} = (ad - bc) \cdot I \rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = (1/(ad - bc)) \cdot \begin{bmatrix} d & b \\ c & a \end{bmatrix} when ad - bc \neq 0$$

Example: $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$, since $det = 1$. Gauss-Jordan method: $[A|I] \rightarrow [I|A^{-1}]$ using the same $row$ operations on both halves: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{bmatrix}$, that is $\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$. The columns of the inverse: if $A^{-1} = [x_1 ... x_n]$, then $Ax_j = e_j$ — finding the inverse means solving $n$ systems; for $a$ single system $Ax = b$, triangular elimination is usually preferable to explicit computation of $A^{-1}$.

### 3.5.6 $LU$ decomposition and solving systems through factors

$$A = LU : elimination factors — below diagonal of unit lower triangular L,$$
$$elimination result — upper triangular U$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} (verify by multiplication)$$

$$E_k \cdots E_2 E_1 A = U \ \rightarrow \ A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU$$

The inverse of "subtract $a$ multiple of $a row$" is "add that multiple back"; the order of factors is crucial ($the$ right operation acts first). $3 \times 3$ example: $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 4 \\ -2 & 2 & 3 \end{bmatrix}$ with multipliers $m_{21} = 2, m_{31} = -1, m_{32} = 1$ gives $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 2 \end{bmatrix}$ (verify $LU = A row$ by $row$). Solving the system: first $Ly = b$ by forward substitution, then $Ux = y$ by backward substitution; for $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 19 \end{bmatrix}$ we get $y = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and $x =$ [3/2; 2]. $By$ decomposing $A$ once, we can solve systems with many right-hand sides.

### 3.5.7 $LDU$ decomposition, pivots and determinant

$$U = D \cdot (D^{-1}U),$$
$$Dcontainsreferenceelementsof U; A = L \cdot D \cdot (D^{-1}U)withunitdiagonalsforLand$$

$$D^{-1}U$$

$$\begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \cdot [1\frac{1}{2};$$
$$01]$$

Pivots and determinant: for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a \neq 0$, the pivots are $a$ and $d-$ cb$/a$, and their product is ad − bc $= detA$. Without $row$ exchanges, the determinant equals the product of pivots; the normalized $LDU$ is unique when it exists with nonzero pivots.

### 3.5.8 Permutations, $PA = LU$ and partial pivoting

Why exchange rows: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ is invertible, but its first pivot is zero; exchanging rows gives $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$ — $a$ zero pivot may indicate $a$ poor $row$ order, not singularity. Permutation matrices permute rows of $I$; there is exactly one 1 in each $row$ and each column. $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ exchanges rows: $PA$ — rows of $A, AP$ — columns of $A$; there are $n!$ permutations of $n$ rows. Example: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ maps $(x_1, x_2, x_3)^T$ to $(x_2, x_3, x_1)^T$ and preserves vector lengths and scalar products.

$$P^{-1} = P^T(sinceP^T \cdot P = I); withexchangesPA = LU$$

Example: $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = I, U = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; with exchanges, it's important to correctly track the multipliers in $L$. Partial pivoting: $[10^{-8}1; 1\ 1]$ is nonsingular, but the tiny first pivot gives $a$ multiplier of $10^8$ — exchanging rows avoids this; the rule is to take the largest in magnitude element of the active column (below and including the pivot).

### 3.5.9 Transposition, symmetric matrices, $LDL^T$

$$(A^T)_{ij} = a_{ji}, (A + B)^T = A^T + B^T, (AB)^T = B^T \cdot A^T$$

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$. $A$ symmetric matrix satisfies $A^T = A$. $Ifa$ symmetric

$A$ has an $LDU$ decomposition without exchanges, its normalized upper factor equals $L^T$: $A = LDL^T$; for example $\begin{bmatrix}2 & 2\\ 2 & 5\end{bmatrix} = \begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix} \cdot \begin{bmatrix}2 & 0\\ 0 & 3\end{bmatrix} \cdot \begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$.

### 3.5.10 Applications: differences, sums, tridiagonal systems

The difference matrix $A = \begin{bmatrix}1 & 0 & 0\\ -1 & 1 & 0\\ 0 & -1 & 1\end{bmatrix}$ gives $Ax = (x_1, x_2 - x_1, x_3 - x_2)^T$; its inverse $A^{-1} = \begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\end{bmatrix}$ gives cumulative sums — the discrete analog of differentiation and integration. The tridiagonal $A = \begin{bmatrix}2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 2\end{bmatrix}$ describes nearest neighbor connections (finite difference equations): its factors preserve the narrow band, and solving via $LU$ uses the structure, avoiding the dense inverse matrix.

### 3.5.11 Week's Summary

- • Matrix multiplication has several equivalent interpretations, but is

non-commutative.

- • The inverse exists only for non-singular square matrices. • Elimination is encoded in $A = LU = LDU$. • Row exchanges are encoded by the matrix $P$: $PA = LU$. • Transpositions and permutations prepare for the vector spaces of week 7.

## 3.2 Linear Algebra I, week 6: In-class problems — questions andfull solutions

[Source: Week 6 Problems in class (Solved) — 20 problems with solutions, from $2 \times 2$ to advanced decompositions]

**Exercise 1.** $2 \times 2$ **inverse. Find** $A^{-1}$ **for** $A = \begin{bmatrix}2 & 1\\ 2 & 3\end{bmatrix}$**. Verify by multiplication.**

Solution. $detA = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. $By$ the $2 \times 2$ formula: $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix}3 & -1\\ -2 & 2\end{bmatrix}$ = [3/4 −1/4; −1/2 1/2]. Verification: $A \cdot A^{-1} = \begin{bmatrix}2 & 1\\ 2 & 3\end{bmatrix}$ ·[3/4 −1/4; −1/2 1/2] = [$\frac{3}{2} - \frac{1}{2}, -\frac{1}{2} + \frac{1}{2}$; $\frac{3}{2} - \frac{3}{2}, -\frac{1}{2} + \frac{3}{2}$] = $I$.

**Exercise 2.** $2 \times 2$ **system solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}, b = \begin{bmatrix}8\\ 6\end{bmatrix}$**.**

Solution. System: 3x + 2y $= 8, x+$ 4y $= 6$. Multiplying the second equation by 3 and subtracting the first, we get 10y $= 10$, that is, $y = 1, x = 6 - 4 = 2$. Thus, $x = \begin{bmatrix}2\\1\end{bmatrix}$.

Verification: $A \cdot x = [3 \cdot 2 + 2;\ 2 + 4] = \begin{bmatrix}8\\6\end{bmatrix} = b$.

**Exercise 3.** $2 \times 2$ **inverse. Find** $A^{-1}$ **for** $A = \begin{bmatrix}4 & 1\\2 & 5\end{bmatrix}$**. Verify by multiplication.**

Solution. $detA = 4 \cdot 5 - 1 \cdot 2 = 18 \neq 0$. $A^{-1} = (\frac{1}{18}) \cdot \begin{bmatrix}5 & -1\\-2 & 4\end{bmatrix}$ = [5/18 −1/18; −1/9 2/9].

Verification: $A \cdot A^{-1} = I$ — for example, element (1,1) equals $4 \cdot (\frac{5}{18}) - \frac{1}{9} = \frac{20}{18} - \frac{2}{18} = 1$, element (2,2) equals $-\frac{2}{18} + \frac{20}{18} = 1$, off-diagonal elements are zeros.

**Exercise 4. Solving** $a2 \times 2$ **system. Solve** $Ax = b$**, where** $A = \begin{bmatrix}2 & 2\\1 & 6\end{bmatrix}, b = \begin{bmatrix}4\\7\end{bmatrix}$**.**

Solution. System: 2x + 2y $= 4$(that is, $x + y = 2), x+$ 6y $= 7$. Subtracting: 5y $= 5, y = 1, x = 1$. Thus, $x = \begin{bmatrix}1\\1\end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix}2 & + & 2\\1 & + & 6\end{bmatrix} = \begin{bmatrix}4\\7\end{bmatrix} = b$.

**Exercise 5.** $2 \times 2$ **decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}2 & 1\\2 & 4\end{bmatrix}$**.**

Solution. Pivot element 2, multiplier $m_{21} = \frac{2}{2} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix}2 & 1\\0 & 3\end{bmatrix}$. $We$ write the multiplier in $L$: $L = \begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}, A = LU$(verification: $\begin{bmatrix}2 & 1\end{bmatrix} + \begin{bmatrix}0 & 3\end{bmatrix} = \begin{bmatrix}2 & 4\end{bmatrix}$). $We$ take the pivot elements 2 and 3 to $D = \begin{bmatrix}2 & 0\\0 & 3\end{bmatrix}$; normalizing rows of $U$ by dividing by pivot elements, we get $D^{-1}U$ = [1 1/2; 0 1]. Thus, $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 6.** $2 \times 2$ **decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}3 & 1\\3 & 5\end{bmatrix}$**.**

Solution. $m_{21} = \frac{3}{3} = 1$: $R_2 \leftarrow R_2 - R_1$ gives $U = \begin{bmatrix}3 & 1\\0 & 4\end{bmatrix}$. $L = \begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}, A = LU$(verification: $\begin{bmatrix}3 & 1\end{bmatrix} + \begin{bmatrix}0 & 4\end{bmatrix} = \begin{bmatrix}3 & 5\end{bmatrix}$). $D = \begin{bmatrix}3 & 0\\0 & 4\end{bmatrix}, D^{-1}U$ = [1 1/3; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 7.** $2 \times 2$ **decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}1 & 1\\4 & 3\end{bmatrix}$**.**

Solution. $m_{21} = 4$: $R_2 \leftarrow R_2 - 4R_1$ gives $U = \begin{bmatrix}1 & 1\\0 & -1\end{bmatrix}$. $L = \begin{bmatrix}1 & 0\\4 & 1\end{bmatrix}, A = LU$(verification: $4 \cdot \begin{bmatrix}1 & 1\end{bmatrix} + \begin{bmatrix}0 & -1\end{bmatrix} = \begin{bmatrix}4 & 3\end{bmatrix}$). The second pivot element is negative — this is acceptable: $D = \begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}, D^{-1}U = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$, and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 8.** $2 \times 2$ **decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

Solution. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1$ gives $U =$ [2 1; 0 3/2] $(4 - \frac{5}{2} = \frac{3}{2})$. $L =$ [1 0; 5/2 1], $A = LU$(verification: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} +$ [0 3/2] $= \begin{bmatrix} 5 & 4 \end{bmatrix}$). $D =$ [2 0; 0 3/2], $D^{-1}U =$ [1 1/2; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 9.** $3 \times 3$ **elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

Solution. Augmented matrix $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 2|0)$. Step $2(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2| - 2)$. Triangular system: $x+$ 3y $+z = 6, -y = -2,$ 2z $= -2$. Back substitution: $z = -1, y = 2, x = 6-6+1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

**Exercise 10.** $3 \times 3$ **elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. Step $2(m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3| - 3)$. Triangular system: $x + y + z = 2, y = 2,$ 3z $= -3$. Back substitution: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

**Exercise 11.** $3 \times 3$ **Elimination (degenerate case). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — the second equation is exactly twice the first; $R_3 - R_1 = (0, 1, 4| - 2)$. After swapping rows 2 and 3: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — two pivots for three unknowns, one free variable. Let $z = t$: $y = -2-$ 4t (from $y+$ 4z $= -2$), $x = 4-$ 2y $-z = 8+$ 7t. General

solution $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix}$, $t \in R$ — infinitely many solutions; the solution given in the file $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ is $a$ particular solution at $t = -1$. Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$. Geometrically: the first two equations define one plane, its intersection with the third is $a$ line.

**Exercise 12.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3 | -3)$. Triangular system: $x + 3y + z = 6, -y = -2, 3z = -3$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

**Exercise 13.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. Multipliers $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot is $2, m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. Thus, $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$(third $row$: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, normalized upper multiplier $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 14.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot is $3, m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; verification: third $row$ of

$LU$ is $-\begin{bmatrix}2 & 1 & 1\end{bmatrix} + 2\cdot\begin{bmatrix}0 & 3 & 1\end{bmatrix} + \begin{bmatrix}0 & 0 & 2\end{bmatrix} = \begin{bmatrix}-2 & 5 & 3\end{bmatrix}$. $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 2\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 15.** $LU$ **and** $LDU3\times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix}2 & 1 & 1\\ 2 & 5 & 2\\ -2 & 11 & 5\end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0,4,1)$; $R_3 + R_1 = (0,12,6)$. Second pivot is $4, m_{32} = 3$: $R_3 - 3R_2 = (0,0,3)$. $L = \begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ -1 & 3 & 1\end{bmatrix}$, $U = \begin{bmatrix}2 & 1 & 1\\ 0 & 4 & 1\\ 0 & 0 & 3\end{bmatrix}$; verification: $-\begin{bmatrix}2 & 1 & 1\end{bmatrix} + 3\cdot\begin{bmatrix}0 & 4 & 1\end{bmatrix} + \begin{bmatrix}0 & 0 & 3\end{bmatrix} = \begin{bmatrix}-2 & 11 & 5\end{bmatrix}$. $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 3\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 16.** $LU$ **and** $LDU3\times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix}2 & 1 & 1\\ 4 & 4 & 3\\ -2 & 1 & 1\end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0,2,1)$; $R_3 + R_1 = (0,2,2)$. Second pivot is $2, m_{32} = 1$: $R_3 - R_2 = (0,0,1)$. $L = \begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ -1 & 1 & 1\end{bmatrix}$, $U = \begin{bmatrix}2 & 1 & 1\\ 0 & 2 & 1\\ 0 & 0 & 1\end{bmatrix}$; verification: second $row$ is $2\cdot\begin{bmatrix}2 & 1 & 1\end{bmatrix} + \begin{bmatrix}0 & 2 & 1\end{bmatrix} = \begin{bmatrix}4 & 4 & 3\end{bmatrix}$, third is $-\begin{bmatrix}2 & 1 & 1\end{bmatrix} + \begin{bmatrix}0 & 2 & 1\end{bmatrix} + \begin{bmatrix}0 & 0 & 1\end{bmatrix} = \begin{bmatrix}-2 & 1 & 1\end{bmatrix}$. $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 17. Row Permutation and** $PLU$**. Find the** $row$ **permutation** $P$ **and factors satisfying** $PA = LU$**, for** $A = \begin{bmatrix}0 & 2 & 1\\ 1 & 1 & 0\\ 2 & 3 & 2\end{bmatrix}$**.**

Solution. The first diagonal element is zero — swap rows 1 and 2: $P = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$, $PA = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix}$. Elimination: element (2, 1) is already zero ($m_{21} = 0$); $m_{31} = 2$: $R_3 - 2R_1 = (2,3,2) - 2\cdot(1,1,0) = (0,1,2)$; then $m_{32} = \frac{1}{2}$: $R_3 - (\frac{1}{2})R_2 = (0,1,2) - (0,1,\frac{1}{2}) = (0,0,\frac{3}{2})$. Thus, $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]. Check: $LU = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix} = PA$(third

$row$: $2\cdot(1,1,0)+(\frac{1}{2})\cdot(0,2,1)+(0,0,\frac{3}{2})=(2,3,2)$). Thus, $PA=LU$: $P$ reorders the equations, giving $a$ suitable pivot; $L$ contains the multipliers 2 and 1/2 with their original signs, although the operations subtract them.

**Exercise 18. Transpose and inverse. For** $A=\begin{bmatrix}1&2&0\\0&1&1\\2&0&1\end{bmatrix}$ **compute** $A^T$ **and verify** $(A^T)^{-1}=(A^{-1})^T$**.**

Solution. Transpose: $A^T=\begin{bmatrix}1&0&2\\2&1&0\\0&1&1\end{bmatrix}$. $detA=1\cdot(1\cdot1-1\cdot0)-2\cdot(0\cdot1-1\cdot2)+0=1+4=5$.

$By$ the adjugate matrix $(A^T)^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&2&-2\\-2&1&4\\2&-1&1\end{bmatrix}$. Independently compute $A^{-1}=(\frac{1}{5})\cdot\begin{bmatrix}1&-2&2\\2&1&-1\\-2&4&1\end{bmatrix}$; then $(A^{-1})^T=(\frac{1}{5})\cdot\begin{bmatrix}1&2&-2\\-2&1&4\\2&-1&1\end{bmatrix}=(A^T)^{-1}$. The equality holds; check: $A^T\cdot(A^T)^{-1}=I$.

**Exercise 19. Symmetric** $LDL^T$**. Factor the symmetric** $A=\begin{bmatrix}2&2&0\\2&5&1\\0&1&3\end{bmatrix}$ **as** $LDL^T$**.**

Solution. Pivot element 2; $m_{21}=\frac{2}{2}=1(m_{31}=0)$: $R_2-R_1=(0,3,1)$. Second pivot element 3; $m_{32}=\frac{1}{3}$: $R_3-(\frac{1}{3})R_2=(0,1,3)-(0,1,\frac{1}{3})=(0,0,\frac{8}{3})$. For symmetric $A$ without exchanges, the normalized upper multiplier is $L^T$, that is, $A=LDL^T$, where $L=$ [1 0 0; 1 1 0; 0 1/3 1], $D=$ [2 0 0; 0 3 0; 0 0 8/3]. Check: $L\cdot D=$ [2 0 0; 2 3 0; 0 1 8/3], and $(LD)\cdot L^T=\begin{bmatrix}2&2&0\\2&5&1\\0&1&3\end{bmatrix}=A$.

**Exercise 20. Challenge problem: tridiagonal model. Solve** $Ax=b$ **using** $LU$ **for** $A=\begin{bmatrix}2&-1&0\\-1&2&-1\\0&-1&2\end{bmatrix}, b=\begin{bmatrix}2\\0\\2\end{bmatrix}$**. Interpret the nearest neighbor structure.**

Solution. Factorization: $m_{21}=-\frac{1}{2}$, so $R_2\leftarrow R_2+(\frac{1}{2})R_1$: $(-1,2,-1)+(1,-\frac{1}{2},0)=(0,\frac{3}{2},-1)$; $m_{32}=-\frac{2}{3}$, so $R_3\leftarrow R_3+(\frac{2}{3})R_2$: $(0,-1,2)+(0,1,-\frac{2}{3})=(0,0,\frac{4}{3})$. Thus, $L=$ [1 0 0; −1/2 1 0; 0 −2/3 1], $U=$ [2 −1 0; 0 3/2 −1; 0 0 4/3]. Forward substitution $Ly=b$: $y_1=2$; $-(\frac{1}{2})\cdot2+y_2=0$, so $y_2=1$; $-(\frac{2}{3})\cdot1+y_3=2$, so $y_3=\frac{8}{3}$. Back substitution $Ux=y$: $(\frac{4}{3})x_3=\frac{8}{3}, x_3=2$; $(\frac{3}{2})x_2-2=1, x_2=2$; $2x_1-2=2, x_1=2$. Thus, $x=\begin{bmatrix}2\\2\\2\end{bmatrix}$. Check: $A\cdot x=\begin{bmatrix}4&&2\\-2&+&4&2\\-2&+&4\end{bmatrix}=\begin{bmatrix}2\\0\\2\end{bmatrix}=b$. The matrix is tridiagonal — nonzero elements only on the

main diagonal and adjacent positions: each unknown is connected only to its nearest neighbors.

## 3.3 Linear Algebra I, Week 6: Practical Session — Questions andFull Solutions

[Source: Week 6 Practical Session exercises with Solutions — 20 problems; Selected detailed solutions — detailed versions of
problems 9–12, 17, 20]

**Exercise 1.** $2 \times 2$ **inverse. Find** $A^{-1}$ **for** $A = \begin{bmatrix} 3 & 2 \\ 1 & 6 \end{bmatrix}$**. Verify by multiplication.**

Solution. $detA = 3 \cdot 6 - 2 \cdot 1 = 16 \neq 0$. $A^{-1} = (\frac{1}{16}) \cdot \begin{bmatrix} 6 & -2 \\ -1 & 3 \end{bmatrix}$ = [3/8 −1/8; −1/16 3/16]. Check: $A \cdot A^{-1} = I$ — element (1,1) is $\frac{9}{8} - \frac{2}{16} = 1$, element (2,2) is $-\frac{1}{8} + \frac{18}{16} = 1$, off-diagonal elements are zero.

**Exercise 2.** $2 \times 2$ **system solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 14 \\ 12 \end{bmatrix}$**.**

Solution. System: 4x $+y = 14$, 2x + 3y $= 12$. Multiplying the first equation by 3 and subtracting the second, we get 10x $= 30$, that is, $x = 3, y = 14 - 12 = 2$. Thus, $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Check: $A \cdot x = \begin{bmatrix} 12 & + & 2 \\ 6 & + & 6 \end{bmatrix} = \begin{bmatrix} 14 \\ 12 \end{bmatrix} = b$.

**Exercise 3.** $2 \times 2$ **inverse. Find** $A^{-1}$ **for** $A = \begin{bmatrix} 2 & 2 \\ 1 & 4 \end{bmatrix}$**. Verify by multiplication.**

Solution. $detA = 2 \cdot 4 - 2 \cdot 1 = 6 \neq 0$. $A^{-1} = (\frac{1}{6}) \cdot \begin{bmatrix} 4 & -2 \\ -1 & 2 \end{bmatrix}$ = [2/3 −1/3; −1/6 1/3]. Check: $A \cdot A^{-1} = I$ — element (1,1) is $\frac{4}{3} - \frac{1}{3} = 1$, element (2,2) is $-\frac{1}{3} + \frac{4}{3} = 1$, off-diagonal elements are zero.

**Exercise 4.** $2 \times 2$ **system solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 14 \end{bmatrix}$**.**

Solution. System: 3x $+y = 8$, 2x + 5y $= 14$. Multiplying the first equation by 5 and subtracting the second, we get 13x $= 26$, that is, $x = 2, y = 8 - 6 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Check: $A \cdot x = \begin{bmatrix} 6 & + & 2 \\ 4 & + & 10 \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \end{bmatrix} = b$.

**Exercise 5.** $2 \times 2$ **factorization. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix} 3 & 1 \\ 5 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{5}{3}$: $R_2 \leftarrow R_2 - (\frac{5}{3})R_1$ gives $U$ = [3 1; 0 10/3] $(5 - 5 = \frac{10}{3})$. $L$ = [1 0; 5/3 1], $A = LU$(check: $(\frac{5}{3}) \cdot \begin{bmatrix} 3 & 1 \end{bmatrix}$ + [0 10/3] $= \begin{bmatrix} 5 & 5 \end{bmatrix}$). $D$ = [3 0; 0 10/3], $D^{-1}U$ = [1 1/3; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 6.** $2 \times 2$ **factorization. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ gives $U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $A = LU$(check: $2 \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix}$). Both pivots are 1, so $D = I$ and $D^{-1}U = U$: $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

**Exercise 7.** $2 \times 2$ **Factorization. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{3}{2}$: $R_2 \leftarrow R_2 - (\frac{3}{2})R_1$ gives $U =$ [2 1; 0 5/2] $(4 - \frac{3}{2} = \frac{5}{2})$. $L =$ [1 0; 3/2 1], $A = LU$(check: $(\frac{3}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} +$ [0 5/2] $= \begin{bmatrix} 3 & 4 \end{bmatrix}$). $D =$ [2 0; 0 5/2], $D^{-1}U =$ [1 1/2; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 8.** $2 \times 2$ **Factorization. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix} 3 & 1 \\ 4 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = \frac{4}{3}$: $R_2 \leftarrow R_2 - (\frac{4}{3})R_1$ gives $U =$ [3 1; 0 11/3] $(5 - \frac{4}{3} = \frac{11}{3})$. $L =$ [1 0; 4/3 1], $A = LU$(check: $(\frac{4}{3}) \cdot \begin{bmatrix} 3 & 1 \end{bmatrix} +$ [0 11/3] $= \begin{bmatrix} 4 & 5 \end{bmatrix}$). $D =$ [3 0; 0 11/3], $D^{-1}U =$ [1 1/3; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 9.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$**.**

Solution. System: $x + y + z = 2$, 2x + 3y + 2z $= 6, x +$ 2y + 4z $= 1$; augmented matrix $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. Eliminate $x(m_{21} = 2, m_{31} = 1)$: new second $row(2, 3, 2|6) - 2 \cdot (1, 1, 1|2) = (0, 1, 0|2)$; new third $row(1, 2, 4|1) - (1, 1, 1|2) = (0, 1, 3|-1)$. Eliminate $y(m_{32} = 1)$: $(0, 1, 3| - 1) - (0, 1, 0|2) = (0, 0, 3| - 3)$. Triangular system: $x + y + z = 2, y = 2$, 3z $= -3$. Back substitution: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Check: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

**Exercise 10.** $3 \times 3$ **Elimination (infinitely many solutions). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$**.**

Solution. System: $x +$ 2y $+ z = 4$, 2x + 4y + 2z $= 8, x +$ 3y + 5z $= 2$; augmented matrix $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{bmatrix}$. Eliminate $x(m_{21} = 2, m_{31} = 1)$: second $row$ gives (0, 0, 0 | 0) — equation

$0 = 0$ imposes no restrictions (second equation is exactly twice the first); third $row$: (0, 1, 4 | −2). Swapping rows 2 and 3, we get echelon form $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — two pivots for three unknowns. Let $z = t, t \in R$: from second $rowy = -2 - 4t$; from first $x = 4 - 2y - z = 8 + 7t$. All solutions: $x = \begin{bmatrix} 8 + 7t \\ -2 - 4t \\ t \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + t \cdot \begin{bmatrix} 7 \\ -4 \\ 1 \end{bmatrix}$, $t \in R$. Check: $(8 + 7t) + 2(-2 - 4t) + t = 4$ and $(8 + 7t) + 3(-2 - 4t) + 5t = 2$; second equation is automatically satisfied as twice the first. Geometrically: first two equations describe the same plane, and its intersection with the third is $a$ line (solution family). Short answer file $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ — particular solution at $t = -1$: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$.

**Exercise 11.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

Solution. System: $x + 3y + z = 6$, $2x + 5y + 2z = 10, x + 4y + 4z = 5$; augmented matrix $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Eliminate $x (m_{21} = 2, m_{31} = 1)$: second $row(0, -1, 0 | -2)$; third $row(0, 1, 3 | -1)$. Second pivot −1; to eliminate the element below it, add $row2$ to $row3 (m_{32} = -1)$: $(0, 1, 3 | -1) + (0, -1, 0 | -2) = (0, 0, 3 | -3)$. Triangular system: $x + 3y + z = 6, -y = -2$, $3z = -3$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Check: $A \cdot x = \begin{bmatrix} 1 + 6 & 1 \\ 2 + 10 & 2 \\ 1 + 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

**Exercise 12.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$.

Solution. System: $x + y + z = 2$, $2x + 3y + 2z = 6, x + 2y + 3z = 2$; augmented matrix $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. Eliminate $x (m_{21} = 2, m_{31} = 1)$: second $row(0, 1, 0 | 2)$; third $row(0, 1, 2 | 0)$. Eliminate $y (m_{32} = 1)$: $(0, 1, 2 | 0) - (0, 1, 0 | 2) = (0, 0, 2 | -2)$. Triangular system: $x + y + z = 2, y = 2$, $2z = -2$. Back substitution: $z = -1, y = 2, x = 2 - 2 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

Check: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} = b.$

**Exercise 13.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot element $3, m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; check: third $row$ of $LU$ is $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 14.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0, 4, 1)$; $R_3 + R_1 = (0, 12, 6)$. Second pivot element $4, m_{32} = 3$: $R_3 - 3R_2 = (0, 0, 3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; check: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 11 & 5 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 15.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot element $2, m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; check: $2 \cdot \begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 3 \end{bmatrix}$ and $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 16.** $LU$ **and** $LDU 3 \times 3$**. Find** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 2 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot element $3, m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; check: third $row$ of $LU$ is $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 17. Row Permutations and $PLU$. Find the $row$ permutation $P$ and factors satisfying $PA = LU$ for $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 2 \end{bmatrix}$.**

Solution. (1) First diagonal element 0 — cannot use it as pivot; we swap rows 1 and 2. The corresponding permutation matrix (same swap in $I$): $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; multiplying $A$ by $P$ on the left swaps rows: $PA = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix}$. (2) Eliminate below first pivot element 1: element (2, 1) is already zero ($m_{21} = 0$); $m_{31} = \frac{2}{1} = 2$, and $R_3 \leftarrow R_3 - 2R_1$: $(2, 3, 2) - 2 \cdot (1, 1, 0) = (0, 1, 2)$. (3) Second pivot element 2; $m_{32} = \frac{1}{2}$, and $R_3 \leftarrow R_3 - (\frac{1}{2})R_2$: $(0, 1, 2) - (0, 1, \frac{1}{2}) = (0, 0, \frac{3}{2})$; $U =$ [1 1 0; 0 2 1; 0 0 3/2]. (4) Build $L$ from multipliers: $L =$ [1 0 0; $m_{21}$10; $m_{31}m_{32}$1] = [1 0 0; 0 1 0; 2 1/2 1]. (5) Check: $LU = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 3 & 2 \end{bmatrix} = PA$ — third $row$ is $2 \cdot (1, 1, 0) + (\frac{1}{2}) \cdot (0, 2, 1) + (0, 0, \frac{3}{2}) = (2, 3, 2)$. Thus $PA = LU$: $P$ reorders equations to provide $a$ suitable pivot, and elimination completes the factorization; $L$ contains the multipliers 2 and 1/2 with their original signs, although the operations subtract them.

**Exercise 18. Transpose and Inverse. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$ compute $A^T$ and verify $(A^T)^{-1} = (A^{-1})^T$.**

Solution. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. $detA = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 \cdot (0 \cdot 0 - 1 \cdot 2) = 5$. From the adjugate matrix $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Independently: $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, from which $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. The equality holds;

check: $A^T \cdot (A^T)^{-1} = I$.

**Exercise 19. Symmetric $LDL^T$. Factor the symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as $LDL^T$.**

Solution. Pivot element 2; $m_{21} = 1 (m_{31} = 0)$: $R_2 - R_1 = (0, 3, 1)$. Second pivot element 3; $m_{32} = \frac{1}{3}$: $R_3 - (\frac{1}{3})R_2 = (0, 0, \frac{8}{3})$. For symmetric $A$ without exchanges $A = LDL^T$, where $L =$ [1 0 0; 1 1 0; 0 1/3 1], $D =$ [2 0 0; 0 3 0; 0 0 8/3]. Check: $L \cdot D =$ [2 0 0; 2 3 0; 0 1 8/3], and $(LD) \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

**Exercise 20. Challenge problem: tridiagonal model. Solve $Ax = b$ using $LU$ for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.**

Solution. (1) Decomposition: $m_{21} = (-1)/2 = -\frac{1}{2}$, so $R_2 \leftarrow R_2 + (\frac{1}{2})R_1$: $(-1, 2, -1) + (1, -\frac{1}{2}, 0) = (0, \frac{3}{2}, -1)$; $m_{32} = (-1)/(\frac{3}{2}) = -\frac{2}{3}$, so $R_3 \leftarrow R_3 + (\frac{2}{3})R_2$: $(0, -1, 2) + (0, 1, -\frac{2}{3}) = (0, 0, \frac{4}{3})$. Thus, $L =$ [1 0 0; −1/2 1 0; 0 −2/3 1], $U =$ [2 −1 0; 0 3/2 −1; 0 0 4/3]; the multipliers appear in $L$ with their original signs, including minuses. (2) Forward substitution $Ly = b$: $y_1 = 3$; $-(\frac{1}{2}) \cdot 3 + y_2 = 0, y_2 = \frac{3}{2}$; $-(\frac{2}{3}) \cdot (\frac{3}{2}) + y_3 = 2, y_3 = 3$. (3) Back substitution $Ux = y$: $(\frac{4}{3})x_3 = 3, x_3 = \frac{9}{4}$; $(\frac{3}{2})x_2 - \frac{9}{4} = \frac{3}{2}, x_2 = (\frac{15}{4}) \cdot (\frac{2}{3}) = \frac{5}{2}$; $2x_1 - \frac{5}{2} = 3, x_1 = \frac{11}{4}$. Thus, $x =$ [11/4; 5/2; 9/4]. Verification: $2 \cdot (\frac{11}{4}) - \frac{5}{2} = 3$; $-(\frac{11}{4}) + 2 \cdot (\frac{5}{2}) - \frac{9}{4} = 0$; $-(\frac{5}{2}) + 2 \cdot (\frac{9}{4}) = 2$. (4) Nearest neighbor structure: the matrix is tridiagonal (nonzero elements only on the main diagonal and immediately above and below it); $x_2$ interacts with both $x_1$ and $x_3$, there is no direct $x_1$–$x_3$ connection $(a_{13} = a_{31} = 0)$. Such $a$ structure arises in heat conduction models along $a$ rod, spring chains, and one-dimensional finite difference equations ($Aisthe$ standard discrete negative second derivative up to $a$ grid step factor); elimination preserves the narrow structure ($LandUare$ bidiagonal), so the system is solved cheaply. $(5) LDU$: $D =$ [2 0 0; 0 3/2 0; 0 0 4/3] and $D^{-1}U =$ [1 −1/2 0; 0 1 −2/3; 0 0 1], so $A = L \cdot D \cdot D^{-1}U$.

## 3.4 Linear Algebra I, Week 6: ISA exercises — questions and fullsolutions

[Source: Week 6 Independent Study Assignment — 40 assigned problems (from basic $2 \times 2$ to challenge problems); instructions:

show elimination steps, verify products, and justify existence of inverse or decomposition. Solutions verified in Python]

**Exercise 1. $2 \times 2$ Inverse. Find $A^{-1}$ for $A = \begin{bmatrix} 2 & 2 \\ 1 & 6 \end{bmatrix}$. Verify by multiplication.**

Solution. $detA = 2 \cdot 6 - 2 \cdot 1 = 10 \neq 0$. $A^{-1} = (\frac{1}{10}) \cdot \begin{bmatrix} 6 & -2 \\ -1 & 2 \end{bmatrix} =$ [3/5 −1/5; −1/10 1/5]. Verification: $A \cdot A^{-1} = I$ — element (1,1) is $\frac{6}{5} - \frac{1}{5} = 1$, element (2,2) is $-\frac{1}{5} + \frac{6}{5} = 1$, off-diagonal elements are zeros.

**Exercise 2. $2 \times 2$ System Solution. Solve $Ax = b$, where $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$.**

Solution. System: 3x $+y = 8$, 2x + 3y $= 10$. Multiplying the first equation by 3 and subtracting the second, we get 7x $= 14$, so $x = 2, y = (8-6)/1 = 2$. Thus, $x = \begin{bmatrix}2\\2\end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix}6 & + & 2\\4 & + & 6\end{bmatrix} = \begin{bmatrix}8\\10\end{bmatrix} = b$.

**Exercise 3.** $2 \times 2$ **Inverse. Find** $A^{-1}$ **for** $A = \begin{bmatrix}4 & 2\\1 & 4\end{bmatrix}$**. Verify by multiplication.**

Solution. $detA = 4 \cdot 4 - 2 \cdot 1 = 14 \neq 0$. $A^{-1} = (\frac{1}{14}) \cdot \begin{bmatrix}4 & -2\\-1 & 4\end{bmatrix}$ = [2/7 −1/7; −1/14 2/7]. Verification: $A \cdot A^{-1} = I$ — element (1,1) is $\frac{8}{7} - \frac{1}{7} = 1$, element (2,2) is $-\frac{1}{7} + \frac{8}{7} = 1$, off-diagonal elements are zeros.

**Exercise 4.** $2 \times 2$ **System Solution. Solve** $Ax = b$**, where** $A = \begin{bmatrix}2 & 1\\2 & 5\end{bmatrix}, b = \begin{bmatrix}4\\12\end{bmatrix}$**.**

Solution. System: 2x $+y = 4$, 2x + 5y $= 12$. Subtracting the first equation from the second: 4y $= 8, y = 2$; then 2x $= 4 - 2, x = 1$. Thus, $x = \begin{bmatrix}1\\2\end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix}2 & + & 2\\2 & + & 10\end{bmatrix} = \begin{bmatrix}4\\12\end{bmatrix} = b$.

**Exercise 5.** $2 \times 2$ **Decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}2 & 1\\5 & 4\end{bmatrix}$**.**

Solution. $m_{21} = \frac{5}{2}$: $R_2 \leftarrow R_2 - (\frac{5}{2})R_1$ gives $U$ = [2 1; 0 3/2]. $L$ = [1 0; 5/2 1], $A = LU$(verification: $(\frac{5}{2}) \cdot \begin{bmatrix}2 & 1\end{bmatrix}$ + [0 3/2] = $\begin{bmatrix}5 & 4\end{bmatrix}$). $D$ = [2 0; 0 3/2], $D^{-1}U$ = [1 1/2; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 6.** $2 \times 2$ **Decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}3 & 1\\2 & 5\end{bmatrix}$**.**

Solution. $m_{21} = \frac{2}{3}$: $R_2 \leftarrow R_2 - (\frac{2}{3})R_1$ gives $U$ = [3 1; 0 13/3] $(5 - \frac{2}{3} = \frac{13}{3})$. $L$ = [1 0; 2/3 1], $A = LU$(verification: $(\frac{2}{3}) \cdot \begin{bmatrix}3 & 1\end{bmatrix}$ + [0 13/3] = $\begin{bmatrix}2 & 5\end{bmatrix}$). $D$ = [3 0; 0 13/3], $D^{-1}U$ = [1 1/3; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 7.** $2 \times 2$ **Decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}1 & 1\\3 & 5\end{bmatrix}$**.**

Solution. $m_{21} = 3$: $R_2 \leftarrow R_2 - 3R_1$ gives $U = \begin{bmatrix}1 & 1\\0 & 2\end{bmatrix}$. $L = \begin{bmatrix}1 & 0\\3 & 1\end{bmatrix}, A = LU$(verification: $3 \cdot \begin{bmatrix}1 & 1\end{bmatrix} + \begin{bmatrix}0 & 2\end{bmatrix} = \begin{bmatrix}3 & 5\end{bmatrix}$). $D = \begin{bmatrix}1 & 0\\0 & 2\end{bmatrix}$, $D^{-1}U = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}$, and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 8.** $2 \times 2$ **Decomposition. Find** $A = LU = LDU$ **for** $A = \begin{bmatrix}2 & 1\\4 & 4\end{bmatrix}$**.**

Solution. $m_{21} = 2$: $R_2 \leftarrow R_2 - 2R_1$ gives $U = \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}$. $L = \begin{bmatrix}1 & 0\\2 & 1\end{bmatrix}, A = LU$(verification:

$2 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $D^{-1}U =$ [1 1/2; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 9.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3 | -3)$. The triangular system: $x +$ 3y $+ z = 6, -y = -2,$ 3z $= -3$; hence $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

**Exercise 10.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0 | 2)$; $R_3 - R_1 = (0, 1, 2 | 0)$. Step $2 (m_{32} = 1)$: $R_3 - R_2 = (0, 0, 2 | -2)$. The triangular system: $x + y + z = 2, y = 2,$ 2z $= -2$; hence $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 2 & 1 \\ 2 & + & 6 & 2 \\ 1 & + & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} = b$.

**Exercise 11.** $3 \times 3$ **Elimination (infinitely many solutions). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 6 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 6 & 1 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0 | 0)$ — the second equation is twice the first and adds no constraints; $R_3 - R_1 = (0, 1, 5 | -3)$. After swapping rows 2 and 3: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 5 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ — two pivots for three unknowns. Free variable $z = t$: $y = -3 -$ 5t, $x = 4 -$ 2y $- z = 10 +$ 9t. Infinitely many

solutions: $x = \begin{bmatrix} 10 & + & 9t \\ -3 & & 5t \\ t & & \end{bmatrix}, t \in R$; particular solution $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ corresponds to $t = -1$.

Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & + & 4 & 1 \\ 2 & + & 8 & 2 \\ 1 & + & 6 & 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 1 \end{bmatrix} = b.$

**Exercise 12.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 2 | 0)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2 | -2)$. Back substitution: $z = -1, y = 2, x = 6 - 6 + 1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 1 & + & 6 & 1 \\ 2 & + & 10 & 2 \\ 1 & + & 8 & 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b.$

**Exercise 13.** $LU$ **and** $LDU 3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 4 & 3 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$. Second pivot $2, m_{32} = 1$: $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$(third $row$: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 1 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 14.** $LU$ **and** $LDU 3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 4 & 2 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1$: $R_2 - R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$. Second pivot $3, m_{32} = 2$: $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; verification: third $row$ of $LU$ is

$-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+2\cdot\begin{bmatrix}0 & 3 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 2\end{bmatrix}=\begin{bmatrix}-2 & 5 & 3\end{bmatrix}$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 2\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 15.** $LU$ **and** $LDU3\times 3$**. Compute** $LU$ **and** $LDU$ **for** $A=\begin{bmatrix}2 & 1 & 1\\ 4 & 6 & 3\\ -2 & 11 & 5\end{bmatrix}$**.**

Solution. $m_{21}=2, m_{31}=-1$: $R_2-2R_1=(0,4,1)$; $R_3+R_1=(0,12,6)$. Second pivot $4, m_{32}=3$: $R_3-3R_2=(0,0,3)$. $L=\begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ -1 & 3 & 1\end{bmatrix}$, $U=\begin{bmatrix}2 & 1 & 1\\ 0 & 4 & 1\\ 0 & 0 & 3\end{bmatrix}$; verification: $-\begin{bmatrix}2 & 1 & 1\end{bmatrix}+3\cdot\begin{bmatrix}0 & 4 & 1\end{bmatrix}+\begin{bmatrix}0 & 0 & 3\end{bmatrix}=\begin{bmatrix}-2 & 11 & 5\end{bmatrix}$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 3\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 16.** $LU$ **and** $LDU3\times 3$**. Compute** $LU$ **and** $LDU$ **for** $A=\begin{bmatrix}2 & 1 & 1\\ 2 & 3 & 2\\ -2 & 1 & 1\end{bmatrix}$**.**

Solution. $m_{21}=1, m_{31}=-1$: $R_2-R_1=(0,2,1)$; $R_3+R_1=(0,2,2)$. Second pivot $2, m_{32}=1$: $R_3-R_2=(0,0,1)$. $L=\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ -1 & 1 & 1\end{bmatrix}$, $U=\begin{bmatrix}2 & 1 & 1\\ 0 & 2 & 1\\ 0 & 0 & 1\end{bmatrix}$; verification: $LU=A$. $D=\begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L\cdot D\cdot D^{-1}U$.

**Exercise 17. Row Permutation and** $PLU$**. Find** $arow$ **permutation** $P$ **and factors satisfying** $PA = LU$**, for** $A=\begin{bmatrix}0 & 2 & 1\\ 1 & 1 & 0\\ 2 & 3 & 2\end{bmatrix}$**.**

Solution. Zero first pivot — swap rows 1 and 2: $P=\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$, $PA=\begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix}$. Elimination: $m_{31}=2(R_3-2R_1=(0,1,2))$; $m_{32}=\frac{1}{2}(R_3-(\frac{1}{2})R_2=(0,0,\frac{3}{2}))$. $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]; verification: $LU=\begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix}=PA$. Thus, $PA = LU$.

**Exercise 18. Transpose and Inverse. For** $A=\begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 2 & 0 & 1\end{bmatrix}$ **compute** $A^T$ **and verify**

$(A^T)^{-1} = (A^{-1})^T$.

Solution. $A^T = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$; $detA = 1\cdot(1-0) - 2\cdot(0-2) + 0 = 5$. $(A^T)^{-1} = (\frac{1}{5})\cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$. Independently: $A^{-1} = (\frac{1}{5})\cdot \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$, from which $(A^{-1})^T = (\frac{1}{5})\cdot \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix} = (A^T)^{-1}$. The equality holds; verification: $A^T \cdot (A^T)^{-1} = I$.

**Exercise 19. Symmetric $LDL^T$. Factor the symmetric $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ as $LDL^T$.**

Solution. The multipliers are $m_{21} = 1, m_{31} = 0, m32 = \frac{1}{3}$. $L$ = [1 0 0; 1 1 0; 0 1/3 1], $D$ = [2 0 0; 0 3 0; 0 0 8/3]; for $a$ symmetric matrix without pivoting, $A = LDL^T$. Verification: $L \cdot D \cdot L^T = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix} = A$.

**Exercise 20. Challenge problem: tridiagonal model. Solve $Ax = b$ using $LU$ for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.**

Solution. $LU$: $m_{21} = -\frac{1}{2}$ and $m_{32} = -\frac{2}{3}$, so $L$ = [1 0 0; −1/2 1 0; 0 −2/3 1], $U$ = [2 −1 0; 0 3/2 −1; 0 0 4/3]. Forward substitution $Ly = b$: $y_1 = 2, y_2 = 1, y_3 = \frac{8}{3}$. Back substitution $Ux = y$: $x_3 = 2, x_2 = 2, x_1 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 4 & & 2 \\ -2 & + & 4 & 2 \\ -2 & + & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$. The matrix is tridiagonal: connections only between nearest neighbors ($x_2$ with $x_1$ and $x_3$; no direct connection $x_1$–$x_3, a_{13} = a_{31} = 0$); $L$ and $U$ are bidiagonal — the narrow structure is preserved.

**Exercise 21. $2 \times 2$ inverse. Find $A^{-1}$ for $A = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$. Verify by multiplication.**

Solution. $detA = 2 \cdot 3 - 1 \cdot 2 = 4 \neq 0$. $A^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix}$ = [3/4 −1/4; −1/2 1/2]. Verification: $A \cdot A^{-1} = \begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix}$ ·[3/4 −1/4; −1/2 1/2] $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

**Exercise 22. Solving $a2 \times 2$ system. Solve $Ax = b$, where $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$.**

Solution. System: 3x + 2y = $8, x+$ 4y = 6. From the second equation $x = 6-$ 4y;

substitution: $3(6-4y)+$ 2y $=8$, that is 18 − 10y $=8, y=1, x=2$. Thus, $x=\begin{bmatrix}2\\1\end{bmatrix}$.

Verification: $A\cdot x=\begin{bmatrix}6&+&2\\2&+&4\end{bmatrix}=\begin{bmatrix}8\\6\end{bmatrix}=b$.

**Exercise 23.** $2\times 2$ **inverse. Find** $A^{-1}$ **for** $A=\begin{bmatrix}4&1\\2&5\end{bmatrix}$**. Verify by multiplication.**

Solution. $detA=4\cdot 5-1\cdot 2=18\neq 0$. $A^{-1}=(\frac{1}{18})\cdot\begin{bmatrix}5&-1\\-2&4\end{bmatrix}$ = [5/18 −1/18; −1/9 2/9]. Verification: $A\cdot A^{-1}=I$ — element (1,1) equals $\frac{20}{18}-\frac{2}{18}=1$, element (2,2) equals $-\frac{2}{18}+\frac{20}{18}=1$, off-diagonal elements are zeros.

**Exercise 24. Solving** $a2\times 2$ **system. Solve** $Ax=b$**, where** $A=\begin{bmatrix}2&2\\1&6\end{bmatrix}, b=\begin{bmatrix}4\\7\end{bmatrix}$**.**

Solution. System: 2x + 2y $=4$(that is, $x+y=2), x+$ 6y $=7$. Subtracting: 5y $=5, y=1, x=1$. Thus, $x=\begin{bmatrix}1\\1\end{bmatrix}$. Verification: $A\cdot x=\begin{bmatrix}2&+&2\\1&+&6\end{bmatrix}=\begin{bmatrix}4\\7\end{bmatrix}=b$.

**Exercise 25.** $2\times 2$ **Factorization. Find** $A=LU=LDU$ **for** $A=\begin{bmatrix}2&1\\2&4\end{bmatrix}$**.**

Solution. $m_{21}=1$: $R_2\leftarrow R_2-R_1$ gives $U=\begin{bmatrix}2&1\\0&3\end{bmatrix}$. $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}, A=LU$(verification: $\begin{bmatrix}2&1\end{bmatrix}+\begin{bmatrix}0&3\end{bmatrix}=\begin{bmatrix}2&4\end{bmatrix}$). $D=\begin{bmatrix}2&0\\0&3\end{bmatrix}, D^{-1}U=$ [1 1/2; 0 1], and $A=L\cdot D\cdot D^{-1}U$.

**Exercise 26.** $2\times 2$ **Factorization. Find** $A=LU=LDU$ **for** $A=\begin{bmatrix}3&1\\3&5\end{bmatrix}$**.**

Solution. $m_{21}=1$: $R_2\leftarrow R_2-R_1$ gives $U=\begin{bmatrix}3&1\\0&4\end{bmatrix}$. $L=\begin{bmatrix}1&0\\1&1\end{bmatrix}, A=LU$(verification: $\begin{bmatrix}3&1\end{bmatrix}+\begin{bmatrix}0&4\end{bmatrix}=\begin{bmatrix}3&5\end{bmatrix}$). $D=\begin{bmatrix}3&0\\0&4\end{bmatrix}, D^{-1}U=$ [1 1/3; 0 1], and $A=L\cdot D\cdot D^{-1}U$.

**Exercise 27.** $2\times 2$ **Factorization. Find** $A=LU=LDU$ **for** $A=\begin{bmatrix}1&1\\4&3\end{bmatrix}$**.**

Solution. $m_{21}=4$: $R_2\leftarrow R_2-4R_1$ gives $U=\begin{bmatrix}1&1\\0&-1\end{bmatrix}$. $L=\begin{bmatrix}1&0\\4&1\end{bmatrix}, A=LU$(verification: $4\cdot\begin{bmatrix}1&1\end{bmatrix}+\begin{bmatrix}0&-1\end{bmatrix}=\begin{bmatrix}4&3\end{bmatrix}$). $D=\begin{bmatrix}1&0\\0&-1\end{bmatrix}$ (negative pivot is acceptable), $D^{-1}U=\begin{bmatrix}1&1\\0&1\end{bmatrix}$, and $A=L\cdot D\cdot D^{-1}U$.

**Exercise 28.** $2\times 2$ **Factorization. Find** $A=LU=LDU$ **for** $A=\begin{bmatrix}2&1\\5&4\end{bmatrix}$**.**

Solution. $m_{21}=\frac{5}{2}$: $R_2\leftarrow R_2-(\frac{5}{2})R_1$ gives $U=$ [2 1; 0 3/2]. $L=$ [1 0; 5/2 1], $A=$

$LU$(verification: $(\frac{5}{2}) \cdot \begin{bmatrix} 2 & 1 \end{bmatrix}$ + [0 3/2] = $\begin{bmatrix} 5 & 4 \end{bmatrix}$). $D$ = [2 0; 0 3/2], $D^{-1}U$ = [1 1/2; 0 1], and $A = L \cdot D \cdot D^{-1}U$.

**Exercise 29.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 3 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix}$.

Solution. Augmented matrix $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 3 & 6 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0| - 2)$; $R_3 - R_1 = (0, 1, 2|0)$. Step $2(m_{32} = -1)$: $R_3 + R_2 = (0, 0, 2| - 2)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 6 \\ 10 \\ 6 \end{bmatrix} = b$.

**Exercise 30.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 4 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 2 & 6 \\ 1 & 2 & 4 & 1 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 1, 0|2)$; $R_3 - R_1 = (0, 1, 3| - 1)$. Step $2(m_{32} = 1)$: $R_3 - R_2 = (0, 0, 3| - 3)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 \\ 6 \\ 1 \end{bmatrix} = b$.

**Exercise 31.** $3 \times 3$ **Elimination (infinitely many solutions). Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 3 & 5 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix}$.

Solution. The augmented matrix is $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 3 & 5 & 2 \end{bmatrix}$. Step $1(m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, 0, 0|0)$ — the second equation is twice the first; $R_3 - R_1 = (0, 1, 4| - 2)$. After swapping rows 2 and 3: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & 4 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. Free variable $z = t$: $y = -2 -$ 4t, $x = 4 -$ 2y $- z = 8 +$ 7t.

Infinitely many solutions: $x = \begin{bmatrix} 8 & + & 7t \\ -2 & & 4t \\ t & & \end{bmatrix}, t \in R$; particular solution $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ ($t =$

$-1$). Verification: $A \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} = b$.

**Exercise 32.** $3 \times 3$ **Elimination. Solve** $Ax = b$ **by Gaussian elimination for** $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 5 & 2 \\ 1 & 4 & 4 \end{bmatrix}, b = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$**.**

Solution. The augmented matrix is $\begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 5 & 2 & 10 \\ 1 & 4 & 4 & 5 \end{bmatrix}$. Step $1 (m_{21} = 2, m_{31} = 1)$: $R_2 - 2R_1 = (0, -1, 0 | -2)$; $R_3 - R_1 = (0, 1, 3 | -1)$. Step $2 (m_{32} = -1)$: $R_3 + R_2 = (0, 0, 3 | -3)$. Back substitution: $z = -1, y = 2, x = 1$. Thus, $x = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix} = b$.

**Exercise 33.** $LU$ **and** $LDU 3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ -2 & 1 & 1 \end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1, m_{32} = 1$: $R_2 - R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$; $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; verification: $LU = A$. $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 34.** $LU$ **and** $LDU 3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 5 & 3 \\ -2 & 5 & 3 \end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1, m_{32} = 2$: $R_2 - 2R_1 = (0, 3, 1)$; $R_3 + R_1 = (0, 6, 4)$; $R_3 - 2R_2 = (0, 0, 2)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$; verification: $LU = A$(third $row$: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 0 & 3 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 5 & 3 \end{bmatrix}$). $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}, D^{-1}U =$ [1 1/2 1/2; 0 1 1/3; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 35.** $LU$ **and** $LDU 3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 5 & 2 \\ -2 & 11 & 5 \end{bmatrix}$**.**

Solution. $m_{21} = 1, m_{31} = -1, m_{32} = 3$: $R_2 - R_1 = (0, 4, 1)$; $R_3 + R_1 = (0, 12, 6)$; $R_3 - 3R_2 = (0, 0, 3)$. $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & 3 \end{bmatrix}$; verification: $LU = A$(third $row$: $-\begin{bmatrix} 2 & 1 & 1 \end{bmatrix} + 3 \cdot$

$\begin{bmatrix}0 & 4 & 1\end{bmatrix} + \begin{bmatrix}0 & 0 & 3\end{bmatrix} = \begin{bmatrix}-2 & 11 & 5\end{bmatrix}$). $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 3\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/4; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 36.** $LU$ **and** $LDU3 \times 3$**. Compute** $LU$ **and** $LDU$ **for** $A = \begin{bmatrix}2 & 1 & 1\\ 4 & 4 & 3\\ -2 & 1 & 1\end{bmatrix}$**.**

Solution. $m_{21} = 2, m_{31} = -1, m_{32} = 1$: $R_2 - 2R_1 = (0, 2, 1)$; $R_3 + R_1 = (0, 2, 2)$; $R_3 - R_2 = (0, 0, 1)$. $L = \begin{bmatrix}1 & 0 & 0\\ 2 & 1 & 0\\ -1 & 1 & 1\end{bmatrix}$, $U = \begin{bmatrix}2 & 1 & 1\\ 0 & 2 & 1\\ 0 & 0 & 1\end{bmatrix}$; verification: $LU = A$. $D = \begin{bmatrix}2 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 1\end{bmatrix}$, $D^{-1}U =$ [1 1/2 1/2; 0 1 1/2; 0 0 1]; $A = LU = L \cdot D \cdot D^{-1}U$.

**Exercise 37. Row Permutation and** $PLU$**. Find** $arow$ **permutation** $P$ **and factors satisfying** $PA = LU$**, for** $A = \begin{bmatrix}0 & 2 & 1\\ 1 & 1 & 0\\ 2 & 3 & 2\end{bmatrix}$**.**

Solution. Swap rows 1 and 2 (zero first pivot): $P = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{bmatrix}$, $PA = \begin{bmatrix}1 & 1 & 0\\ 0 & 2 & 1\\ 2 & 3 & 2\end{bmatrix}$. Elimination: $m_{31} = 2(R_3 - 2R_1 = (0, 1, 2))$; $m_{32} = \frac{1}{2}(R_3 - (\frac{1}{2})R_2 = (0, 0, \frac{3}{2}))$. $L =$ [1 0 0; 0 1 0; 2 1/2 1], $U =$ [1 1 0; 0 2 1; 0 0 3/2]; verification: $LU = PA$. Thus, $PA = LU$.

**Exercise 38. Transpose and Inverse. For** $A = \begin{bmatrix}1 & 2 & 0\\ 0 & 1 & 1\\ 2 & 0 & 1\end{bmatrix}$ **compute** $A^T$ **and verify** $(A^T)^{-1} = (A^{-1})^T$**.**

Solution. $A^T = \begin{bmatrix}1 & 0 & 2\\ 2 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$; $detA = 5$. $(A^T)^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix}1 & 2 & -2\\ -2 & 1 & 4\\ 2 & -1 & 1\end{bmatrix}$; on the other hand, $A^{-1} = (\frac{1}{5}) \cdot \begin{bmatrix}1 & -2 & 2\\ 2 & 1 & -1\\ -2 & 4 & 1\end{bmatrix}$, and $(A^{-1})^T = (\frac{1}{5}) \cdot \begin{bmatrix}1 & 2 & -2\\ -2 & 1 & 4\\ 2 & -1 & 1\end{bmatrix} = (A^T)^{-1}$. The equality holds; verification: $A^T \cdot (A^T)^{-1} = I$.

**Exercise 39. Symmetric** $LDL^T$**. Factor the symmetric** $A = \begin{bmatrix}2 & 2 & 0\\ 2 & 5 & 1\\ 0 & 1 & 3\end{bmatrix}$ **as** $LDL^T$**.**

Solution. Multipliers $m_{21} = 1, m_{31} = 0, m_{32} = \frac{1}{3}$. $L =$ [1 0 0; 1 1 0; 0 1/3 1], $D =$ [2 0 0; 0 3 0; 0 0 8/3], and $A = LDL^T$. Verification: $L \cdot D \cdot L^T = \begin{bmatrix}2 & 2 & 0\\ 2 & 5 & 1\\ 0 & 1 & 3\end{bmatrix} = A$.

**Exercise 40. Challenge problem: tridiagonal model. Solve $Ax = b$ using $LU$ for $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$. Interpret the nearest neighbor structure.**

Solution. $LU$: $L = [1\ 0\ 0;\ -1/2\ 1\ 0;\ 0\ -2/3\ 1]$, $U = [2\ -1\ 0;\ 0\ 3/2\ -1;\ 0\ 0\ 4/3]$ ($the$ multipliers $m_{21} = -\frac{1}{2}, m_{32} = -\frac{2}{3}$ are stored in $L$ with their original signs). Forward substitution $Ly = b$ gives $y = [2;\ 1;\ 8/3]$; backward substitution $Ux = y$ gives $x_3 = 2, x_2 = 2, x_1 = 2$. Thus, $x = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$. Verification: $A \cdot x = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = b$.

The matrix is tridiagonal — each unknown is connected only to its nearest neighbors; the factors $L$ and $U$ are bidiagonal, and the narrow structure makes the solution inexpensive.

## 3.5 Linear Algebra I, week 6: review material with examples

[Source: Week 6 Extra material — Review with examples: $I$)$a$ brief lesson on $LU, LDU$, and normalized $U$; $II$)$a$ brief lesson on

Gaussian elimination for $3\times3$ systems; $III$)$a$ detailed example of finding the inverse by the Gauss-Jordan method]

Key points of lesson $II$(review of week 5). $A$ system can be viewed in three pictures: $row$(equations are planes), matrix ($Ax = b$), and column (combination of columns of $A$ gives $b$). Three reversible $row$ operations are allowed: swapping two equations ($R_i \leftrightarrow R_j$), multiplying an equation by $a$ nonzero number ($R_i \leftarrow c \cdot R_i, c \neq 0$), and adding $a$ multiple of another equation ($R_i \leftarrow R_i + c \cdot R_j$); each is applied to the entire equation, including the right-hand side. Procedure for $a3 \times 3$ system: write the augmented matrix; select $a$ nonzero first pivot element (swapping rows if necessary); eliminate elements below it; select the second pivot element in the updated matrix; eliminate the element below it; check for contradictions or free variables; with three pivot elements, perform backward substitution; verify the solution in the original system. Backward substitution formulas: $z = c_3/u_{33}$, $y = (c_2 - u_{23} \cdot z)/u_{22}$, $x = (c_1 - u_{12} \cdot y - u_{13} \cdot z)/u_{11}$(pivot elements are nonzero). Typical errors: forgetting to update the right-hand side; using the old $row$ after it has been changed; confusing the direction of the multiplier relationship; considering $a$ zero proposed pivot element as proof of degeneracy.

**Example 1. $LU$, normalization, and $LDU$(lesson $I$). Find the $LU$ and $LDU$ decompositions of the matrix $A = \begin{bmatrix} 2 & 4 & 2 \\ 4 & 11 & 7 \\ 2 & 7 & 8 \end{bmatrix}$.**

Solution. Eliminate under the first pivot element 2: $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ give $\begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 6 \end{bmatrix}$; then under the second pivot element 3: $R_3 \leftarrow R_3 - R_2$ gives $\begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$.

The multipliers are $m_{21} = 2, m_{31} = 1, m_{32} = 1$, so $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $A = LU$ with $U =$

$\begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$; $L$ stores the multipliers with their original signs ($the$ operation subtracts $2R_1$, but $L_{21} = +2$). Normalization: divide each $row$ of $U$ by its diagonal element — the pivot elements 2, 3, 3 are factored out into $D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, the normalized upper factor is $D^{-1}U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, and $A = L \cdot D \cdot (D^{-1}U)$; multiplication by $D^{-1}$ on the left scales the rows. Note on conventions: in $A = LU$, the pivot elements sit on the diagonal of $U(Disnot$ separated, $L$ has unit diagonal); in $A = LDU$, the pivot elements are in $D$, and both $L$ and the third factor have unit diagonals; many textbooks rename $D^{-1}U$ as $U$.

**Example 2. Solving $a$ system using factors (lesson $I$). Using the decomposition from example 1, solve $Ax = \begin{bmatrix} 8 \\ 22 \\ 17 \end{bmatrix}$.**

Solution. Forward substitution $Ly = b$: $y_1 = 8$; $2y_1 + y_2 = 22$, from which $y_2 = 6$; $y_1 + y_2 + y_3 = 17$, from which $y_3 = 3$, that is, $y = \begin{bmatrix} 8 \\ 6 \\ 3 \end{bmatrix}$. Backward substitution $Ux = y$: $3x_3 = 3, x_3 = 1$; $3x_2 + 3x_3 = 6, x_2 = 1$; $2x_1 + 4x_2 + 2x_3 = 8, x_1 = 1$. Thus, $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. With $LDU$, the computation breaks into three steps: $Ly = b, Dz = y$(division by pivot elements), then $(D^{-1}U)x = z$. When $row$ exchanges are needed, the standard form is $PA = LU$ or $PA = LDU$, where $P$ records the exchanges; extraction and division by pivot elements requires them to be nonzero.

**Example 3. Complete Gaussian elimination example (lesson $II$). Solve the system $2x + y - z = 1$, $4x + 3y + z = 13$, $-2x + y + 2z = 6$.**

Solution. Here $A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 13 \\ 6 \end{bmatrix}$. Step 1: the first pivot element is 2, multiplier $m_{21} = \frac{4}{2} = 2$; $R_2 \leftarrow R_2 - 2R_1$: $(4x + 3y + z) - 2(2x + y - z) = 13 - 2 \cdot 1$, that is, $y + 3z = 11$. Step 2: $m_{31} = -\frac{2}{2} = -1$; $R_3 \leftarrow R_3 + R_1$: $(-2x + y + 2z) + (2x + y - z) = 6 + 1$, that is, $2y + z = 7$. Step 3: the second pivot element is 1 (coefficient of $y$ in the updated equation $2), m_{32} = 2$; $R_3 \leftarrow R_3 - 2R_2$: $(2y + z) - 2(y + 3z) = 7 - 22$, that is, $-5z = -15$. The pivot elements 2, 1, $-5$ — the second and third are taken from the updated equations, not from the original diagonal. Back substitution: $z = 3, y = 2, x = 1$, that is, $(x, y, z) = (1, 2, 3)$. Verification: $2 \cdot 1 + 2 - 3 = 1$; $4 \cdot 1 + 6 + 3 = 13$; $-2 \cdot 1 + 2 + 6 = 6$. With the augmented matrix: $\begin{bmatrix} 2 & 1 & -1 & 1 \\ 4 & 3 & 1 & 13 \\ -2 & 1 & 2 & 6 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 2 & 1 & 7 \end{bmatrix} \to \begin{bmatrix} 2 & 1 & -1 & 1 \\ 0 & 1 & 3 & 11 \\ 0 & 0 & -5 & -15 \end{bmatrix}$.

**Example 4. Elimination as matrix multiplication (Lesson $II$). Write the operations of Example 3 in terms of elimination matrices.**

Solution. For $R_2 \leftarrow R_2 - 2R_1$: $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ($the$ second $row$ of the product $E_{21} \cdot A$ equals $-2 \cdot (row1 of A) + (row2 of A)$); for adding $row1$ to $row3$: $E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$; for $R_3 \leftarrow R_3 - 2R_2$: $E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. Complete elimination: $E_{32} \cdot E_{31} \cdot E_{21} \cdot A = U$ and $E_{32} \cdot E_{31} \cdot E_{21} \cdot b = c$, where $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 11 \\ -15 \end{bmatrix}$; the rightmost operation acts first. Rules: $(BC)_{ij} = \sum_k b_{ik} \cdot c_{kj}$; associativity $(BC)D = B(CD)$; non-commutativity $BC \neq CB$ — therefore the order of $row$ operations is important.

**Example 5. Zero proposed pivot element (Lesson $II$). Solve the system $y + z = 5, x + y + z = 6$, 2x + 3y + 4z = 20.**

Solution. $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 5 \\ 6 \\ 20 \end{bmatrix}$; the first proposed pivot element is zero — we swap rows 1 and 2 (permutation matrix $P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, reordered system PAx = $Pb$): $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 1 & 5 \\ 2 & 3 & 4 & 20 \end{bmatrix}$. Then $R_3 \leftarrow R_3 - 2R_1$: (0, 1, 2 | 8); then $R_3 \leftarrow R_3 - R_2$: (0, 0, 1 | 3). Back substitution: $z = 3, y = 2, x = 1$. $A$ zero in the proposed pivot position does not automatically mean no unique solution; for numerical calculations, partial pivoting takes the largest available element in absolute value in the current column, controlling the elimination multipliers.

**Example 6. How many solutions does the system have (Lesson $II$). Consider the family $x + y + z = 6$, 2x + 3y + 4z = 20, 3x + 4y + 5z = $k$. For which $k$ are there no solutions, for which are there infinitely many?**

Solution. Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$: we get $y +$ 2z $= 8$ and $y +$ 2z $= k - 18$. Subtracting equation 2 from equation 3: $0 = k - 26$. $If k \neq 26$ — no solutions (when $k = 27$ we get $0 = 1$ — the equations are inconsistent). $If k = 26$ — the last equation is $0 = 0$ and adds no constraints: with free variable $z = t$ we have $y = 8 -$ 2t and $x = 6 - y - z = t - 2$, that is, $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$ — infinitely many solutions. Outcome table for $3 \times 3$: three pivot elements — unique solution; $a row$ [000|$d$], $d \neq 0$ — no solutions; fewer than three pivot elements without $a$ contradictory $row$ — infinitely many solutions ($one$ zero $row$ by itself does not guarantee consistency — need to check the other rows).

**Example 7.** $LU$**-factorization and reuse (Lesson** $II$**). Write** $A = LU$ **for the system of Example 3 and solve** $Lc = b, Ux = c$**.**

Solution. The multipliers of Example 3: $m_{21} = 2, m_{31} = -1, m_{32} = 2$; we place them under the diagonal: $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$, and $A = LU$ with $U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix}$; the signs in $L$ are the multipliers themselves, while the elimination operations use their negatives. Solve in two steps: $Lc = b$ gives $c_1 = 1, c_2 = 11, c_3 = -15$(forward substitution); then $Ux = c$ by back substitution gives (1, 2, 3). The advantage appears when several systems have the same matrix $A$ but different right-hand sides: $A$ is factored once, and $L$ and $U$ are reused. $If row$ exchanges are needed, the standard factorization is $PA = LU$.

**Example 8. Exercises** $A$ **and** $B$(**Lesson** $II$**). Solve the systems:** $(A)x + y + z = 6$**, 2x + 3y** $+z = 11, x+$ **2y + 3z** $= 14$**;** $(B)y+$ **2z** $= 8, x + y + z = 6$**, 2x + 3y** $+z = 11$**.**

Solution. $(A)$ Eliminate $x$: $R_2 - 2R_1$ gives $y - z = -1$; $R_3 - R_1$ gives $y+$ 2z $= 8$. Subtracting: 3z $= 9, z = 3, y = z - 1 = 2, x = 6 - 2 - 3 = 1$. Answer (1, 2, 3); check: $1+2+3 = 6, 2+6+3 = 11, 1+4+9 = 14$. $(B)$ There is no $x$ in the first equation (zero proposed pivot element) — before eliminating $x$, swap the first two rows: $x + y + z = 6, y+$ 2z $= 8$, 2x + 3y $+z = 11$. Eliminate $x$: $R_3 - 2R_1$ gives $y - z = -1$; together with $y+$ 2z $= 8$, subtraction gives 3z $= 9, z = 3, y = 2, x = 1$. Answer (1, 2, 3); check: $2 + 2 \cdot 3 = 8, 1 + 2 + 3 = 6, 2 + 6 + 3 = 11$.

**Example 9. Exercises** $C$ **and** $D$(**Lesson** $II$**).** $(C)$ **Determine for which** $k$ **the system** $x+y+z = 6$**, 2x + 3y + 4z** $= 20$**, 3x + 4y + 5z** $= k$ **has no solutions, and for which it has infinitely many.** $(D)$ **Find the matrix of the operation** $R_3 \leftarrow R_3 - 4R_1$**.**

Solution. $(C)$ This is $a$ family of example 6: after elimination, the third equation reduces to $0 = k - 26$, so when $k \neq 26$ there are no solutions, when $k = 26$ there are infinitely many: $(x, y, z) = (t - 2, 8 - 2t, t), t \in R$; there is no value of $k$ that gives $a$ unique solution ($the$ coefficient matrix has only two pivots). $(D)$ The operation changes only $row3$, subtracting $4 \cdot (row1)$: $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$; the third $row$ of the product $E \cdot A$ is $(row3 of A) - 4 \cdot (row1 of A)$, the other rows do not change.

**Example 10. Inverse matrix by the Gauss-Jordan method (Lesson** $III$**). Find the inverse matrix of** $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$**, by adjoining the identity matrix and reducing the left part to the identity.**

Solution. Theory: the inverse of an $n \times n$ matrix $A$ is the matrix $A^{-1}$ with $AA^{-1} = A^{-1}A = I$; each elementary $row$ operation is equivalent to left multiplication by an elementary matrix, and if $E_k \cdots E_2 E_1 A = I$, then $A^{-1} = E_k \cdots E_2 E_1$ — therefore the same operations, applied to $I$, give $A^{-1}$: $[A|I] \to [I|A^{-1}]$; each operation acts on the entire augmented $row$. Step 1: $[A|I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$. Step 2 (eliminate under the first

pivot): $R_2 \leftarrow R_2 - R_1$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$. Step 3 (under the second): $R_3 \leftarrow R_3 - R_2$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Step 4 (above the third): $R_2 \leftarrow R_2 - R_3$: $\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. Step 5 (above the second): $R_1 \leftarrow R_1 - R_2$: $\begin{bmatrix} 1 & 0 & 0 & 3 & -2 & 1 \\ 0 & 1 & 0 & -2 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{bmatrix}$. The left part became the identity, so $A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. Check by multiplication: $A \cdot A^{-1} = I (for$ example, element (2, 1): $1 \cdot 3 + 2 \cdot (-2) + 1 \cdot 1 = 0)$.

**Example 11. When the method doesn'$t$ work (Lesson $III$). Show that the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has no inverse.**

Solution. $A$ square matrix is invertible only if the left block in $[A|I]$ reduces to the identity matrix. Apply $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{bmatrix}$. $Arow$ of zeros appeared in the left block — there are too few pivots to form $I$; the matrix is degenerate (singular), no inverse exists. Note: Gaussian elimination stops at an upper triangular matrix, while Gauss-Jordan continues until each pivot is 1 and all elements above and below each pivot are zero.

# 4 Week 7 — Transposes, Permutations and VectorSpaces (Linear Algebra I)

## 4.1 Linear Algebra I, week 7: formula catalog

[Source: Week 7 lecture "Transposes, permutations, vector spaces and subspaces" (Strang, Introduction to Linear Algebra, 4th ed.,
§2.7 pp. 108–120, §3.1 pp. 121–132)]

### 4.5.1 Review of Week 6: $A = LU, LDU, PA = LU$

Gaussian elimination gives $E_k \cdots E_1 A = U$. Without $row$ exchanges, the elimination factors combine into $a$ lower triangular $L$(with ones on the diagonal), and $A = LU$: for example, $A = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$, the step $R2 \leftarrow R2 - 2R1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$. Factoring out the pivots gives $A = LD\bar{\text{U}}$: $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ is the diagonal of pivots, and $\bar{\text{U}}$ ($U$ with $a$ bar) = [1 1/2; 0 1] is upper triangular with ones on the diagonal.

$$A = LU, A = LD; whenexchangingrows—PA = LU$$

$If$ the leading element is zero ($A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$), the rows are swapped: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ gives $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; all $row$ swaps are recorded in $P$.

### 4.5.2 Transposition: definition and basic identities

$IfA$ is an $m \times n$ matrix, then $A^T$ has size $n \times m$ and elements $(A^T)_{ij} = a_{ji}$: columns become rows. Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ gives $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$.

$$(A^T)^T = A, (A+B)^T = A^T + B^T, (cA)^T = cA^T$$

Element-wise proof ($forthesum$): $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$.

$$(AB)^T = B^T A^T$$

The order of factors is reversed ($A^T B^T maynot$ even be defined). Proof: $((AB)^T)_{ij} = (AB)_{ji} = \Sigma_k a_{jk} b_{ki} = \Sigma_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. Lecture check: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix}$: $(AB)^T = \begin{bmatrix} 8 & 3 \\ 2 & 1 \end{bmatrix} = B^T A^T$.

$$(A^{-1})^T = (A^T)^{-1}$$

For an invertible $A$, we transpose $AA^{-1} = I$: we get $(A^{-1})^T A^T = I$; similarly $A^T (A^{-1})^T = I$. Thus, transposition and inversion commute.

### 4.5.3 Symmetric and skew-symmetric matrices; the product $A^T A$

$A$ matrix $S$ is symmetric if $S^T = S (s_{ij} = s_{ji})$; example: $S = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$. $A$ matrix $K$ is skew-symmetric if $K^T = -K$; example: $K = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$; the diagonal is zero: $k_{ii} = -k_{ii} \Rightarrow k_{ii} = 0$.

$$A = (A + A^T)/2 + (A - A^T)/2 = S + K, S^T = S, K^T = -K$$

Any square matrix decomposes this way; lecture example: $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \to S = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; verification: $S + K = A$.

$$(A^T A)^T = A^T A, x^T A^T A x = (Ax)^T (Ax) = ||Ax||^2 \geq 0$$

The product $A^T A$ is always symmetric ($a$ special case of $(AB)^T = B^T A^T$), and $x^T A^T A x = ||Ax||^2 \geq 0$ for any $x$. Example: $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ gives $A^T A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$; matrices $R^T R$ are fundamental to least squares. Scalar and outer product: for columns $u, v \in R^n$, the

number $u^T v$ is the scalar product, while $\text{uv}^T$ is an $n \times n$ matrix. Example: $u = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$: $u^T v = 11$, $\text{uv}^T = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$.

### 4.5.4 Permutations: $P^T = P^{-1}$, six $3 \times 3$ matrices, partial pivoting

$A$ permutation matrix is obtained by permuting the rows of the identity matrix: exactly one 1 in each $row$ and each column. Example: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ maps [$x_1$; $x_2$; $x_3$] to [$x_2$; $x_3$; $x_1$].

$$PA—matrix A with permuted rows; AQ—with permuted columns$$

Left multiplication permutes rows: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \to PA = \begin{bmatrix} 3 & 4 \\ 5 & 6 \\ 1 & 2 \end{bmatrix}$. Right multiplication permutes columns: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \to AQ = \begin{bmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{bmatrix}$. $In$ general, $PA \neq AP$.

$$P^T P = I, i.e. P^{-1} = P^T; consequences: ||Px|| = ||x||, detP = \pm 1$$

The columns of $P$ are different coordinate unit vectors, so their scalar products form $I$: $P^T P = I$, hence $P^{-1} = P^T$. Consequences: $||Px|| = ||x||$ and $detP = \pm 1$. One $row$ exchange changes the sign of the determinant: $det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = -1$, so $det(PA) = -detA$; two exchanges restore the sign. There are exactly $n!$ permutations of order $n$; for $n = 3$ there are six (Strang, §2.7) — three even ($detP = +1$) and three odd ($detP = -1$); $P^{-1} = P^T$, therefore $(P_{32}P_{21})^{-1} = P_{21}P_{32}$:

| **Permutation** | **Matrix (row-wise)** | $\det P$ | $P^{-1} = P^T$ |
|---|---|---|---|
| $I$ | $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ | $+1$ | $I$ |
| $P_{21}$ (swap rows 1, 2) | $\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$ | $-1$ | $P_{21}$ |
| $P_{32}$ (swap rows 2, 3) | $\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$ | $-1$ | $P_{32}$ |
| $P_{31}$ (swap rows 1, 3) | $\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$ | $-1$ | $P_{31}$ |
| $P_{32}P_{21}$ (cyclic) | $\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$ | $+1$ | $P_{21}P_{32}$ |
| $P_{21}P_{32}$ (cyclic) | $\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$ | $+1$ | $P_{32}P_{21}$ |

Partial pivoting: if the pivot element is zero, rows are swapped, taking in the current column the candidate with the largest absolute value. Example: $A = [10^{-5}1;\ 1\ 1]$: without swap the multiplier is $\frac{1}{10}^{-5} = 10^5$, after swap — $10^{-5}/1 = 10^{-5}$. Swaps are recorded in $P$: $PA = LU$.

### 4.5.5 Vector spaces: definition and examples

Vector space over $R$ — $a$ set of vectors with addition and scalar multiplication operations, satisfying vector space axioms: zero vector, opposite elements, associativity, distributivity, etc. Lecture examples: $R^n$, real matrices, polynomials, continuous functions — any object can be $a$ "vector".

### 4.5.6 Subspaces and closure testing

$$W \subseteq V — subspace \Leftrightarrow 0 \in W; u + v \in W\, for all u, v \in W; cu \in W\, for all u \in W, c \in R$$

Equivalently: non-empty $W$ is closed under combinations au + bv. Test: (1) zero vector; (2) sum; (3) scalar multiplication — violation of any point refutes.

- • Line through origin: $W = \{(t,\ 2t)\} = span\{(1,\ 2)\}$: for $u = (s,\ 2s), v = (t,\ 2t)$

the combination au + bv $= (as + bt, 2(as + bt))$ is again in $W$ — subspace.

- • Shifted line (counterexample): $S = \{(x, y)\colon\ y = 2x + 1\}$ — zero vector does

not satisfy the equation; $a$ subspace must pass through the origin.

- • Plane through origin: $W = \{(x, y, z)\colon\ x + y + z = 0\} = span\{(1,\ 0,\ -1),\ (0,\ 1,$

$-1)\}$ (from $z = -x - y$: $(x, y, z) = x(1, 0, -1) + y(0, 1, -1)$); plane $x + y + z = 1$ is not $a$ subspace $(no$ zero).

- • Geometry of $R^3$: subspaces are only {0}, lines through 0, planes through 0 and all $R^3$.
- • Intersection and union: $U \cap W$ — subspace; union — not necessarily: $x$ and $y$ axes are subspaces of $R^2$, but $(1,0)+(0,1)=(1,1)$ lies on neither axis.
- • Matrices and polynomials: symmetric $2\times 2$ — subspace of all $2\times 2$(sums and multiples are symmetric); polynomials with $p(0)=0$ — subspace, with $p(0)=1$ — no.

### 4.5.7 Linear combinations and linear spans

$$span v_1, ..., v_k = c_1 v_1 + ... + c_k v_k : c_i \in R$$

Linear $span$ — the set of all combinations of given vectors; it is always $a$ subspace: if $u = \Sigma a_i v_i$ and $v = \Sigma b_i v_i$, then su + tv = $\Sigma(sa_i + tb_i)v_i$ — again $a$ combination. Lecture example: is (3, 5) in $span${(1, 1), (1, −1)}? Solve $c_1 + c_2 = 3, c_1 - c_2 = 5 \to c_1 = 4, c_2 = -1$ — yes.

### 4.5.8 Column space $C(A)$

$$A = [a_1 a_2 ... a_n] \to Ax = x_1 a_1 + ... + x_n a_n;$$
$$C(A) = Col(A) = span a_1, ..., a_n$$

The product $Ax$ is $a$ linear combination of columns of $A$. Column space $C(A)$ = Col$(A)$ — the linear $span$ of columns, $a$ subspace of $R^m$ for an $m \times n$ matrix $A$. Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$ — columns are proportional, Col$(A)$ = $span${(1, 2, 0)} — $a$ line in $R^3$.

$$system Ax = b is solvable \Leftrightarrow b \in C(A)$$

The system $Ax = b$ has $a$ solution if and only if $b$ lies in the column space: the solution $x$ picks the coefficients of the combination of columns that gives $b$. The main criterion for solvability is weak.

### 4.5.9 Null space $N(A)$

$$N(A) = Null(A) = x : Ax = 0$$

Null space — the set of solutions to the homogeneous system; it is $a$ subspace of $R^n$: if $Au = Av = 0$, then $A(su + tv) = 0$. Example: $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$: $x_1 + 2x_2 = 0$ gives $x_1 = -2x_2$, therefore Null$(A)$ = $span${(−2, 1)} — $a$ line.

$$x : Ax = 0 — always subspace; x : Ax = b, b \neq 0 — not subspace(doesn't contain 0)$$

Solutions to the homogeneous system form $a$ subspace; solutions to the non-homogeneous $(b \neq 0)$ — no: zero vector does not belong to them. $If$ the system $Ax = b$ is solvable, its solutions are $a$ shift of the $null$ space: $x = x_p + x_n$, where $x_p$ is $a$ particular solution, $x_n \in$ Null$(A)$(affine set). $By$ week 8 midterm: be able to compute $A^T$ and $(AB)^T$, use $P^{-1} = P^T$ and $PA = LU$, test closure, find spans and membership in $C(A)$.

## 4.2 Linear Algebra I, week 7: in-class problems — questions andfull solutions

[Source: Week 7 Class problems (Solved), Linear Algebra $I$(Fall 2026) — 20 in-class problems; arithmetic verified in python]

**Problem 1. Transpose of $2 \times 2$. Compute $A^T$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.**

Solution. Transpose swaps rows and columns: $(A^T)_{ij} = a_{ji}$. Therefore $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$; in particular, $(A^T)_{12} = a_{21} = 3$, diagonal remains unchanged.

**Problem 2. Transposition of $a$ sum. Compute $(A + B)^T$ for $A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$.**

Solution. First, add: $A + B = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$. Then transpose: $(A + B)^T = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$. Check: $A^T + B^T = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$ — matches.

**Problem 3. Product $A^T A$. Find $A^T A$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.**

Solution. $A^T A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$ — the result is symmetric, as always (exercise $26ISA$).

**Problem 4. Transposition of $a$ product. Find $(AB)^T$ for $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.**

Solution. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$. The identity gives the same: $B^T A^T = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix}$ — the order of factors is reversed.

**Problem 5. Checking symmetry. $Is$ the matrix $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ symmetric?**

Solution. $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$. Yes, it is symmetric: the corresponding off-diagonal elements coincide ($a_{12} = a_{21} = -3$), and the diagonal does not change under transposition.

**Problem 6. Symmetric and skew-symmetric parts. Decompose $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ into symmetric and skew-symmetric parts.**

Solution. $A^T = \begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}$; $S = (A + A^T)/2 = \begin{bmatrix}1 & 3\\ 3 & 3\end{bmatrix}$ ($off-$diagonal elements are averaged), $K = (A - A^T)/2 = \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$. Check: $S + K = A, S^T = S, K^T = -K$.

**Problem 7. Row permutation: $PA$. Find $PA$, where $P = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$.**

Solution. $PA = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} \cdot \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} = \begin{bmatrix}3 & 4\\ 1 & 2\end{bmatrix}$: the first $row$ of the product is the second $row$ of $A$, and vice versa. Left multiplication swaps rows 1 and 2.

**Problem 8. Column permutation: $AP$. Find $AP$ for the same $A$ and $P(A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}, P = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix})$.**

Solution. $AP = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \cdot \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}2 & 1\\ 4 & 3\end{bmatrix}$: the first column of the product is the second column of $A$, and vice versa. Right multiplication swaps columns 1 and 2; note that $AP \neq PA$.

**Problem 9. Inverse of $a$ permutation matrix. Find $P^{-1}$ for $P = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$.**

Solution. For the permutation matrix $P^{-1} = P^T = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$. Verification: $P^T P = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix} \cdot \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix} = I$ — the columns of $P$ are orthonormal.

**Problem 10. Exchange rows 1 and 3. Find the permutation matrix $P$ that exchanges rows 1 and 3 of $a 3 \times 3$ matrix.**

Solution. Exchange of rows 1 and 3, applied to $I_3$: $P = \begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$. For any $A$, the product $PA$ exchanges its first and third rows; for example, for $A = \begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$ we get $\begin{bmatrix}7 & 8 & 9\\ 4 & 5 & 6\\ 1 & 2 & 3\end{bmatrix}$.

**Problem 11. $PA = LU$ with zero pivot. Find $P, L, U$ for $A = \begin{bmatrix}0 & 2\\ 3 & 4\end{bmatrix}$.**

Solution. The pivot element $a_{11} = 0$ — exchange is unavoidable. Exchange the rows: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$. Below the pivot 3 is 0 — elimination is complete, no multipliers: $L = I_2, U = PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; result: $PA = LU$.

**Problem 12. $LU$ without $row$ exchanges. Find $L, U$ for $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ without $row$ exchanges.**

Solution. The pivot element $2 \neq 0$, no exchanges needed; the multiplier $m_{21} = \frac{6}{2} = 3$. Step $R2 \leftarrow R2 - 3R1$: $\begin{bmatrix} 6 & 5 \end{bmatrix} - 3 \cdot \begin{bmatrix} 2 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \end{bmatrix}$, so $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Verification: $LU = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix} = A$.

**Problem 13. Subspace: line through origin. $IsW$ = {$(x, y)$: $y$ = 3x} $a$ subspace of $R^2$?**

Solution. Yes. $W = \{t(1, 3)\colon\ t \in R\}$: the zero vector corresponds to $t = 0$. For $u = (s,$ 3s$), v = (t,$ 3t) and any $a, b$: au + bv $= (as + bt, 3(as + bt)) \in W$ — closure holds; $W$ is $a$ subspace (line through origin).

**Problem 14. Counterexample: shifted line. $IsS$ = {$(x, y)$: $y$ = 3x +2} $a$ subspace?**

Solution. $No$. The zero vector does not satisfy the equation: $3 \cdot 0 + 2 = 2 \neq 0$. The zero vector test is sufficient for refutation: $a$ subspace must contain the origin.

**Problem 15. Subspace: plane. $IsW$ = {$(x, y, z)$: $x + y - z = 0$} $a$ subspace?**

Solution. Yes. For $f(x, y, z) = x + y - z$ and $u, v \in W$: $f(au + bv) = a \cdot f(u) + b \cdot f(v) = 0$, so au + bv $\in W$; and $f(0, 0, 0) = 0$. $W$ is $a$ subspace (plane through origin).

**Problem 16. Linear combination of vectors. Express (4, 1) as $a$ combination of vectors (1, 1) and (1, −1).**

Solution. $We$ seek $c_1(1, 1) + c_2(1, -1) = (4, 1)$: $c_1 + c_2 = 4, c_1 - c_2 = 1$. Adding: $2c_1 = 5, c_1 = \frac{5}{2}, c_2 = \frac{3}{2}$. Verification: $(\frac{5}{2} + \frac{3}{2}, \frac{5}{2} - \frac{3}{2}) = (4, 1)$.

**Problem 17. Span of proportional vectors. Find the linear $span$ of vectors (1, 2, 0) and (2, 4, 0).**

Solution. The second vector is twice the first: $(2, 4, 0) = 2(1, 2, 0)$, so any combination is $a$ multiple of (1, 2, 0). Result: $span${(1, 2, 0), (2, 4, 0)} = $\{t(1, 2, 0)\colon\ t \in R\}$ — $a$ line in $R^3$.

**Problem 18. Null space of $a1 \times 3$ matrix. Find Null$(A)$ for $A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$.**

Solution. Equation $Ax = 0$: $x_1 + 2x_2 - x_3 = 0$. Free variables $x_2 = s, x_3 = t$, then $x_1 = -2$s $+t$ and $x = s(-2, 1, 0) + t(1, 0, 1)$. Result: Null$(A) = span${(−2, 1, 0), (1, 0, 1)} — $a$ plane in $R^3$. Verification: $1 \cdot (-2) + 2 \cdot 1 - 0 = 0$ and $1 \cdot 1 + 2 \cdot 0 - 1 = 0$.

**Problem 19. Column space. Find Col$(A)$ for $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$; does the vector (3, 6, 0) belong to it?**

Solution. Columns $a_1 = (1,2,0)$ and $a_2 = (2,4,0) = 2a_1$ are proportional, so Col$(A) = span\{(1, 2, 0)\}$ — $a$ line in $R^3$. Since $(3,6,0) = 3(1,2,0)$, the vector lies on this line: yes ($for$ example, $A \cdot \begin{bmatrix}3\\0\end{bmatrix} = (3,6,0)$).

**Problem 20. Transpose and determinant of $3\times3$. For $A = \begin{bmatrix}1&2&0\\0&1&1\\0&0&2\end{bmatrix}$ find $A^T$ and $det(A)$.**

Solution. $A^T = \begin{bmatrix}1&0&0\\2&1&0\\0&1&2\end{bmatrix}$. $A$ is upper triangular, so $detA = 1\cdot1\cdot2 = 2$(product of diagonal elements); transposition doesn'$t$ change the determinant: $det(A^T) = 2$.

## 4.3 Linear Algebra I, week 7: ISA exercises — questions andcomplete solutions

[Source: Week 7 Independent Study Assignments, Linear Algebra $I$(Fall 2026) — 50 problems without solutions; solutions created

and verified in python]

Independent Study Assignments — ungraded self-practice (instructions: column vectors, intermediate steps, justification of subspace answers). File parts: $A$(1–12) — transpositions, $B$(13–20) — permutations and pivoting, $C$(21–32) — subspace criteria, $D$(33–42) — linear spans and matrix subspaces, $E$(43–46) — mixed applications, $F$(47–50) — advanced problems. Exercises 1–20 exactly match classroom problems 1–20 (below — compact solutions, complete ones above); the remaining 30 are fully solved.

**Exercise 1. Compute $A^T$ for $A = \begin{bmatrix}1&2\\3&4\end{bmatrix}$.**

Solution. $A^T = \begin{bmatrix}1&3\\2&4\end{bmatrix}$ (rows became columns); in detail — problem 1 above.

**Exercise 2. Compute $(A+B)^T$ for $A = \begin{bmatrix}1&0\\2&3\end{bmatrix}, B = \begin{bmatrix}2&1\\0&-1\end{bmatrix}$.**

Solution. $A+B = \begin{bmatrix}3&1\\2&2\end{bmatrix}$, so $(A+B)^T = \begin{bmatrix}3&2\\1&2\end{bmatrix} = A^T + B^T$.

**Exercise 3. Find $A^TA$ for $A = \begin{bmatrix}1&2\\0&1\end{bmatrix}$.**

Solution. $A^TA = \begin{bmatrix}1&0\\2&1\end{bmatrix}\cdot\begin{bmatrix}1&2\\0&1\end{bmatrix} = \begin{bmatrix}1&2\\2&5\end{bmatrix}$ — the result is symmetric.

**Exercise 4. Find $(AB)^T$ for $A = \begin{bmatrix}1&1\\0&2\end{bmatrix}, B = \begin{bmatrix}2&0\\1&3\end{bmatrix}$.**

Solution. $AB = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}$, so $(AB)^T = \begin{bmatrix} 3 & 2 \\ 3 & 6 \end{bmatrix} = B^T A^T$.

**Exercise 5.** *Is* **the matrix** $A = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix}$ **symmetric?**

Solution. Yes: $A^T = \begin{bmatrix} 2 & -3 \\ -3 & 1 \end{bmatrix} = A$ — the off-diagonal elements coincide ($a_{12} = a_{21} = -3$).

**Exercise 6. Decompose** $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ **into symmetric and skew-symmetric parts.**

Solution. $S = (A + A^T)/2 = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}$, $K = (A - A^T)/2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$; check: $S + K = A$.

**Exercise 7. Find** $PA$**, where** $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$**.**

Solution. $PA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$ — left multiplication swapped rows 1 and 2.

**Exercise 8. Find** $AP$ **for the same** $A$ **and** $P(A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix})$**.**

Solution. $AP = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ — right multiplication swapped columns 1 and 2; note that $AP \neq PA$.

**Exercise 9. Find** $P^{-1}$ **for** $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$**.**

Solution. $P^{-1} = P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; check: $P^T P = I$.

**Exercise 10. Find the permutation matrix** $P$ **that swaps rows 1 and 3 of** $a3 \times 3$ **matrix.**

Solution. $P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ — swapping rows 1 and 3 in $I_3$; $PA$ swaps the first and third rows of any matrix.

**Exercise 11. Find** $P, L, U$ **for** $A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$**.**

Solution. Zero pivot requires exchange: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}$; elimination is already complete, so $L = I_2, U = PA$; result: $PA = LU$.

**Exercise 12. Find $L, U$ for $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \end{bmatrix}$ without $row$ exchanges.**

Solution. $m_{21} = \frac{6}{2} = 3$; step $R2 \leftarrow R2 - 3R1$ gives $U = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$; $L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$; check: $LU = A$.

**Exercise 13.** $IsW$ **= {**$(x, y)$**:** $y =$ **3x} a subspace of** $R^2$**?**

Solution. Yes: $W = \{t(1, 3)\colon t \in R\}$, zero vector when $t = 0$, and $a(s,$ 3s$) + b(t,$ 3t$) = ($as + bt, $3(as + bt)) \in W$.

**Exercise 14.** $IsS$ **= {**$(x, y)$**:** $y =$ **3x +**$2$**} a subspace?**

Solution. $No$: (0, 0) does not satisfy the equation $(0 \neq 3 \cdot 0 + 2)$ — no zero vector.

**Exercise 15.** $IsW$ **= {**$(x, y, z)$**:** $x + y - z = 0$**} a subspace?**

Solution. Yes: $f(x, y, z) = x+y-z$ is linear, $f(0, 0, 0) = 0$ and $f(au+bv) = a \cdot f(u)+b \cdot f(v) = 0$ for $u, v \in W$.

**Exercise 16. Express (4, 1) as** $a$ **combination of vectors (1, 1) and (1, −1).**

Solution. $c_1 + c_2 = 4, c_1 - c_2 = 1 \rightarrow c_1 = \frac{5}{2}, c_2 = \frac{3}{2}$; verification: $(\frac{5}{2})(1, 1) + (\frac{3}{2})(1, -1) = (4, 1)$.

**Exercise 17. Find the linear** $span$ **of vectors (1, 2, 0) and (2, 4, 0).**

Solution. $(2, 4, 0) = 2(1, 2, 0)$, so $span$(1, 2, 0), (2, 4, 0)$\} = \{t(1, 2, 0)\colon t \in R\}$ — $a$ line.

**Exercise 18. Find Null**$(A)$ **for** $A = \begin{bmatrix} 1 & 2 & -1 \end{bmatrix}$**.**

Solution. $x_1 + 2x_2 - x_3 = 0 \rightarrow x_1 = -2$s $+t, x_2 = s, x_3 = t$; Null$(A) = span$(−2, 1, 0), (1, 0, 1)$\}$.

**Exercise 19. Find Col**$(A)$ **for** $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{bmatrix}$**; does (3, 6, 0) belong to it?**

Solution. Columns are proportional: Col$(A) = span$(1, 2, 0)$\}$; since $(3, 6, 0) = 3(1, 2, 0)$, the vector belongs to Col$(A)$.

**Exercise 20. For** $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ **find** $A^T$ **and** $det(A)$**.**

Solution. $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$; $A$ is upper triangular, so $detA = 1 \cdot 1 \cdot 2 = 2 = det(A^T)$.

**Exercise 21. Dimensions. For** $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ **compute** $A^T$ **and specify dimensions.**

Solution. $A$ is $2 \times 3$; $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ is $3 \times 2$: dimensions swap, $(m \times n)^T = n \times m$.

**Exercise 22. Proof** $(sum)$**. Prove** $(A + B)^T = A^T + B^T$ **element-wise.**

Solution. For $A=(a_{ij}), B=(b_{ij})$ of size $m\times n$ and each $i,j$: $((A+B)^T)_{ij}=(A+B)_{ji}=a_{ji}+b_{ji}=(A^T)_{ij}+(B^T)_{ij}=(A^T+B^T)_{ij}$. Elements match — matrices are equal.

**Exercise 23. Proof (product). Prove $(AB)^T=B^TA^T$ for compatible matrices.**

Solution. Let $A$ be $m\times n, B$ be $n\times p$. For each $i,j$: $((AB)^T)_{ij}=(AB)_{ji}=\Sigma_k a_{jk}b_{ki}=\Sigma_k(B^T)_{ik}(A^T)_{kj}=(B^TA^T)_{ij}$. Thus, $(AB)^T=B^TA^T$: the order of factors is reversed.

**Exercise 24. Product $A^TA$ for $3\times 2$. Compute $A^TA$ for $A=\begin{bmatrix}1&0\\2&1\\0&3\end{bmatrix}$ ($in the$ file, the $3\times 2$ matrix is written $row$-wise).**

Solution. $A^TA=\begin{bmatrix}1&2&0\\0&1&3\end{bmatrix}\cdot\begin{bmatrix}1&0\\2&1\\0&3\end{bmatrix}=\begin{bmatrix}5&2\\2&10\end{bmatrix}$ — the result is symmetric, as theory predicts (exercise 26).

**Exercise 25. Symmetric and skew-symmetric parts. For $A=\begin{bmatrix}1&3\\-2&4\end{bmatrix}$ find the symmetric and skew-symmetric parts.**

Solution. $A^T=\begin{bmatrix}1&-2\\3&4\end{bmatrix}$. $S=(A+A^T)/2=$ [1 1/2; 1/2 4], $K=(A-A^T)/2=$ [0 5/2; −5/2 0]. Verification: $S+K=A, S^T=S, K^T=-K$.

**Exercise 26. Symmetry of $A^TA$. Show that $A^TA$ is symmetric for any real matrix $A$.**

Solution. $(A^TA)^T=A^T(A^T)^T=A^TA$(rule $(AB)^T=B^TA^T$): the matrix equals its transpose, $i.e.$, is symmetric ($for A of$ size $m\times n$, the product $A^TA$ is square $n\times n$).

**Exercise 27. Scalar and outer product. Compute $u^Tv$ and uv$^T$ for $u=(2,-1)^T, v=(3,4)^T$.**

Solution. $u^Tv=2\cdot 3+(-1)\cdot 4=2$ — $a$ number (scalar product); uv$^T=\begin{bmatrix}2\\-1\end{bmatrix}\cdot\begin{bmatrix}3&4\end{bmatrix}=\begin{bmatrix}6&8\\-3&-4\end{bmatrix}$ — $a 2\times 2$ matrix (outer product).

**Exercise 28. Permutation of rows 1 and 2. Construct the permutation matrix that exchanges rows 1 and 2 of $I_3$.**

Solution. $P=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$ — $I_3$ with rows 1 and 2 exchanged. For any $A$ with three rows, $PA$ swaps the first two: $P\cdot\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}=\begin{bmatrix}4&5&6\\1&2&3\\7&8&9\end{bmatrix}$.

**Exercise 29. $PA$ vs $AP$. For $P=\begin{bmatrix}0&1\\1&0\end{bmatrix}$ compare $PA$ and $AP$ for $A=\begin{bmatrix}2&5\\7&1\end{bmatrix}$.**

Solution. $PA = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} \cdot \begin{bmatrix}2 & 5\\ 7 & 1\end{bmatrix} = \begin{bmatrix}7 & 1\\ 2 & 5\end{bmatrix}$ — rows are permuted; $AP = \begin{bmatrix}2 & 5\\ 7 & 1\end{bmatrix} \cdot \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}5 & 2\\ 1 & 7\end{bmatrix}$ — columns are permuted. $PA \neq AP$: left multiplication acts on rows, right on columns.

**Exercise 30. Verification $P^TP = I$. For $P = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$ verify $P^TP = I$.**

Solution. $P^T = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}$; $P^TP = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix} \cdot \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix} = I$. Rows of $P^T$ are columns of $P(e_2, e_3, e_1)$, their scalar products give $\delta_{ij}$; consequence: $P^{-1} = P^T$.

**Exercise 31. Determinant of permutation. Find the determinant of the permutation matrix that exchanges rows 1 and 3.**

Solution. $P = \begin{bmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{bmatrix}$; one $row$ exchange changes the sign: $detP = -1$. Verification by expansion along the first $row$: $1 \cdot (-1)^{1+3} \cdot det \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = 1 \cdot 1 \cdot (-1) = -1$.

**Exercise 32. $PA = LU$ for $2 \times 2$. Find $P, L, U$ for $A = \begin{bmatrix}0 & 1\\ 4 & 2\end{bmatrix}$.**

Solution. The pivot $a_{11} = 0$ — exchange is needed: $P = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, $PA = \begin{bmatrix}4 & 2\\ 0 & 1\end{bmatrix}$. Below the pivot 4 is 0, elimination is complete: $L = I_2, U = \begin{bmatrix}4 & 2\\ 0 & 1\end{bmatrix}$; result: $PA = LU$.

**Exercise 33. Partial pivoting. Apply partial pivoting to $A = \begin{bmatrix}0.001 & 1\\ 2 & 1\end{bmatrix}$; compare the multipliers before and after the exchange.**

Solution. Without pivoting: pivot element 0.001, multiplier $m_{21} = \frac{2}{0}.001 = 2000$ — huge (subtracting $2000 \cdot$ (first $row$) would strongly amplify errors). Partial pivoting takes the largest by magnitude candidate in column 1 (number 2); after exchange ($P = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, $PA = \begin{bmatrix}2 & 1\\ 0.001 & 1\end{bmatrix}$) the multiplier $m_{21} = 0.\frac{001}{2} = 0.0005$. The exchange reduced the multiplier from 2000 to $0.0005 (by a$ factor of $4 \cdot 10^6$).

**Exercise 34. Subspace? $x+ 2y = 0$. Check if $W =$ {$(x, y)$: $x+ 2y = 0$} is $a$ subspace.**

Solution. Yes. $(1) 0 + 2 \cdot 0 = 0$ — the zero vector is in $W$. (2) From $x+ 2y = 0$ and $x' + 2y' = 0$ follows $(x + x') + 2(y + y') = 0$. $(3)(cx) + 2(cy) = c(x + 2y) = 0$. Conclusion: $W = span\{(-2, 1)\}$ — $a$ line through the origin.

**Exercise 35. Subspace? $x+ 2y = 3$. Check if $W =$ {$(x, y)$: $x+ 2y = 3$} is $a$ subspace.**

Solution. $No$: $0+2\cdot 0=0\neq 3$ — there is no zero vector in $W$, but $a$ subspace must contain $0(the$ shifted line does not pass through the origin).

**Exercise 36. Subspace? $x\geq 0$. Check if $W=\{(x,y)$: $x\geq 0\}$ is $a$ subspace.**

Solution. $No$. The zero vector is in $W$, and $W$ is closed under addition $(sumofnon-$negative $x$-coordinates is non-negative). But closure under scalar multiplication is violated: $(1,0)\in W$, but $(-1)\cdot(1,0)=(-1,0)\notin W$. $A$ half-plane is not $a$ subspace.

**Exercise 37. Subspace? $x-y+z=0$. Check if $W=\{(x,y,z)$: $x-y+z=0\}$ is $a$ subspace.**

Solution. Yes: $f(x,y,z)=x-y+z$ is linear, $f(0,0,0)=0$ and $f(au+bv)=a\cdot f(u)+b\cdot f(v)=0$ for $u,v\in W$. Explicitly: $x=y-z$ gives $(x,y,z)=y(1,1,0)+z(-1,0,1)$, that is $W=span\{(1,$ 1, 0), (−1, 0, 1)$\}$ — $a$ plane through the origin.

**Exercise 38. Linear combination. Express (6, 2) as $a$ combination of vectors (1, 1) and (1, −1).**

Solution. $c_1+c_2=6, c_1-c_2=2\to c_1=4, c_2=2$. Check: $4(1,1)+2(1,-1)=(6,2)$.

**Exercise 39. Geometry of $span$. Describe geometrically $span\{$(1, 2, 0), (0, 1, 2)$\}$.**

Solution. The vectors are not proportional — the $span$ is two-dimensional: $a$ plane through the origin. Its points: $s(1,2,0)+t(0,1,2)=(s,$ 2s $+t,$ 2t); eliminating parameters $(s=x, t=z/2)$ we get 4x − 2y $+z=0$. Check: $4\cdot 1-2\cdot 2+0=0$ and $4\cdot 0-2\cdot 1+2=0$.

**Exercise 40. Null space of $2\times 3$. Find Null$(A)$ for $A=\begin{bmatrix}1&1&1\\0&1&1\end{bmatrix}$ ($2\times 3$ matrix, written $row$-wise).**

Solution. From $row2$: $x_2=-x_3$; from $row1$: $x_1=-x_2-x_3=0$. Free variable $x_3=t$: $x=t(0,-1,1)$. Conclusion: Null$(A)=span\{$(0, −1, 1)$\}$ — $a$ line in $R^3$. Check: $\begin{bmatrix}1&1&1\\0&1&1\end{bmatrix}\cdot\begin{bmatrix}0\\-1\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$.

**Exercise 41. Column space of $3\times 2$. Find Col$(A)$ for $A=\begin{bmatrix}1&2\\2&4\\3&6\end{bmatrix}$.**

Solution. Columns $a_1=(1,2,3)$ and $a_2=(2,4,6)=2a_1$ are proportional, so any combination is $a$ multiple of $a_1$: Col$(A)=span\{$(1, 2, 3)$\}$ — $a$ line in $R^3$(vectors $(t,$ 2t, 3t)).

**Exercise 42. Membership in column space. Does (1, 1, 1) belong to the column space Col$(A)$ for $A=\begin{bmatrix}1&0\\0&1\\1&1\end{bmatrix}$ ($3\times 2$ matrix)?**

Solution. $We$ seek $c_1(1,0,1)+c_2(0,1,1)=(1,1,1)$: the first two coordinates give $c_1=c_2=1$, but then the third is $2\neq 1$ — contradiction. Conclusion: $(1,1,1)\notin$ Col$(A)(Col(A)isthe$ plane $z=x+y$, and $1\neq 1+1)$.

**Exercise 43. Intersection of subspaces. Prove that the intersection of two subspaces is $a$ subspace.**

Solution. For subspaces $U, W \subseteq V$: $(1) 0 \in U \cap W$. $(2) If u, v \in U \cap W$, then $u + v \in U$ and $u + v \in W$(closure of each), so $u + v \in U \cap W$. $(3) If u \in U \cap W$, then cu $\in U$ and cu $\in W$, so cu $\in U \cap W$. The test is passed.

**Exercise 44. Union of subspaces. Give two subspaces whose union is not $a$ subspace.**

Solution. $In R^2$ take the axes $U = \{(x, 0)\}$ and $W = \{(0, y)\}$ — both are subspaces. $In$ the union lie (1, 0) and (0, 1), but their sum (1, 1) lies on neither axis. The union is not closed under addition — not $a$ subspace.

**Exercise 45. Symmetric $3 \times 3$ matrices. $Is$ the set of symmetric $3 \times 3$ matrices $a$ subspace of the space of all real $3 \times 3$ matrices?**

Solution. Yes. (1) The zero matrix is symmetric. $(2)(S + T)^T = S^T + T^T = S + T$ for symmetric $S, T$. $(3)(cS)^T = c \cdot S^T =$ cS. Closure is satisfied — it is $a$ subspace.

**Exercise 46. Polynomials with $p(0) = 1$. $Is$ the set of polynomials satisfying $p(0) = 1a$ subspace?**

Solution. $No$. The zero polynomial gives $p(0) = 0 \neq 1$ — there is no zero vector. Additionally, for $p(x) = 1$ and $q(x) = 1 + x$: $(p + q)(0) = 2 \neq 1$ — closure under addition is also violated. (Polynomials with $p(0) = 0$, on the other hand, do form $a$ subspace.)

**Exercise 47. Advanced problem. Prove that Null$(A^T A)$ = Null$(A)$ for any real matrix $A$.**

Solution. Two inclusions. $(1) If Ax = 0$, then $A^T Ax = A^T(Ax) = 0$, that is, Null$(A) \subseteq$ Null$(A^T A)$. (2) Let $A^T Ax = 0$; multiply on the left by $x^T$: $x^T A^T Ax = ||Ax||^2 = 0$. The sum of squares of the coordinates of $Ax$ is zero, which means $Ax = 0$. Conclusion: Null$(A^T A)$ = Null$(A)$; $a$ consequence is that $A$ and $A^T A$ have the same $null$ space and $rank$(least squares fact).

**Exercise 48. Advanced problem. $If P$ is $a$ permutation matrix, prove that Col$(PA) = P$ Col$(A)$.**

Solution. $By$ definition, Col$(PA) = \{$PAx: $x \in R^n$ $\}$, $P$ Col$(A) = \{Pb$: $b \in$ Col$(A)\}$. $(1) If y =$ PAx, then $y = P(Ax)$ with $Ax \in$ Col$(A)$ — thus $y \in P$ Col$(A)$. $(2) If y = Pb$ with $b = Ax \in$ Col$(A)$, then $y =$ PAx $\in$ Col$(PA)$. Conclusion: Col$(PA) = P$ Col$(A)$ — permuting rows permutes each point of the column space by the same permutation ($b \to Pbis$ bijective, $P$ is invertible).

**Exercise 49. Advanced problem. $If W$ is $a$ subspace, prove that $span(W) = W$.**

Solution. $span(W)$ is the set of all finite combinations $c_1 w_1 + ... + c_k w_k$ of elements $w_i \in W$. (1) Each combination lies in $W$: $c_i w_i \in W$(scalars), sum of elements of $W$ is in $W$(addition). (2) Each $w \in W$ is the combination $1 \cdot w$.

Conclusion: $span(W) = W$ — $a$ subspace already contains all its linear combinations.

**Exercise 50. Advanced problem. For $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ describe Col$(A)$ and Null$(A)$.**

Solution. Columns $a_1 = (1, 0, 0), a_2 = (2, 1, 0), a_3 = (0, 1, 0) = a_2 - 2a_1$; the third $row$ is zero, so all columns lie in the plane $z = 0$, and $a_1, a_2$ are independent. Result: Col$(A) = span\{$(1, 0, 0), (2, 1, 0)$\} = \{(s + 2t, t, 0)\}$ — the entire plane $z = 0$. Null

space: from $row 2 x_2 = -x_3$, from $row 1 x_1 = -2x_2 = 2x_3$; $x_3 = t$: $x = t(2, -1, 1)$. Result: $\text{Null}(A) = span\{(2, -1, 1)\}$ — $a$ line. Verification: $A \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.